CoE shell model: canceling the leftover CMB residual with added bodies — all scenarios

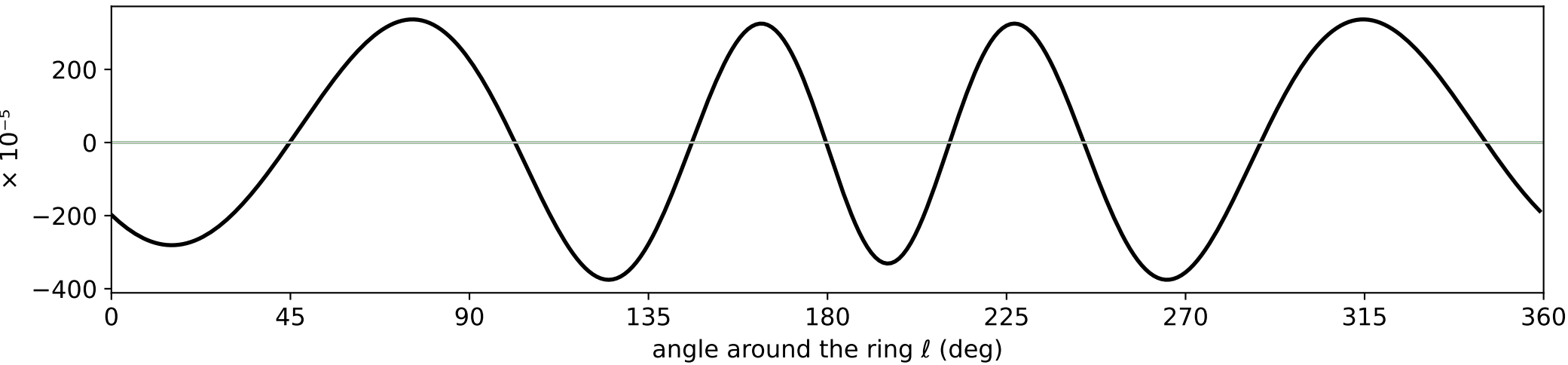
All the scenarios we have run, in one place: leftover vs number of added bodies, an example universe for each (top view), and the residual that remains.

How to read the charts:

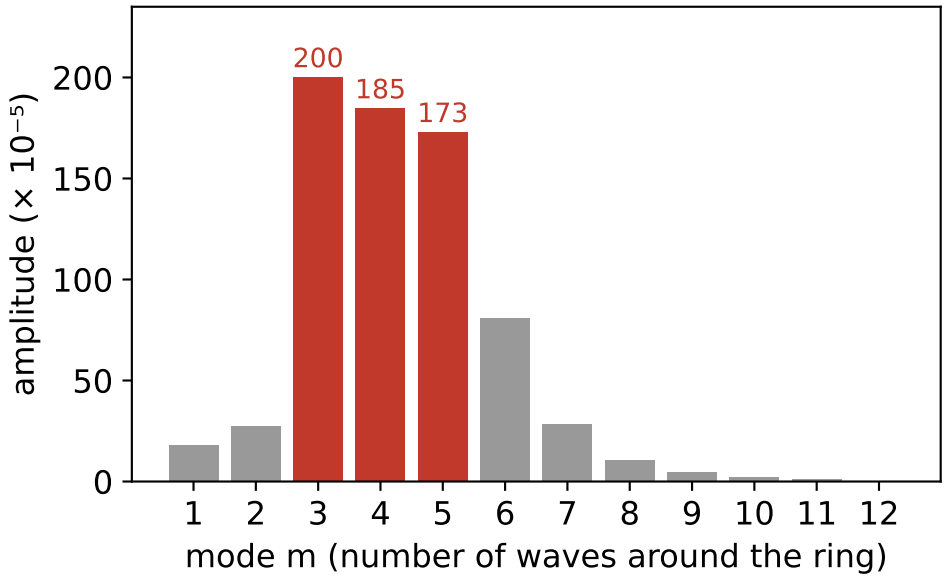
- Leftover = the rms of the CMB pattern that is still unexplained after the fit, in units of the 10^{-5} flatness floor. The green line at 1× is the target ('CMB flat').
- Start = 237×: the leftover after special relativity + the central mass, before any bodies are added. The four clumps reached 89× (grey dotted line).
- Bodies: compact masses (red dots, sized by mass) and deep voids (blue circles, $\delta = -0.7$, drawn at their true radius). 'Inside' = inside our 14.3 Gpc visible sphere; 'outside' = 14.3–43 Gpc from us.
- 'Mode m' = how many waves fit around the ring (m = 1 dipole … m = 12); '13+' collects all finer ripple.
- Top views look down on the galactic plane; bodies above or below it are projected onto it.
- Every example was re-scored from the saved layout with the saved model code; the re-scored leftover matches the saved value in every case (shown at the bottom of each page).
- No new fits were run for this report. Above ~5 bodies, many different layouts fit about equally well (overfitting), so the example layouts are illustrations, not unique answers.

The fingerprint: the leftover CMB pattern every scenario tries to cancel

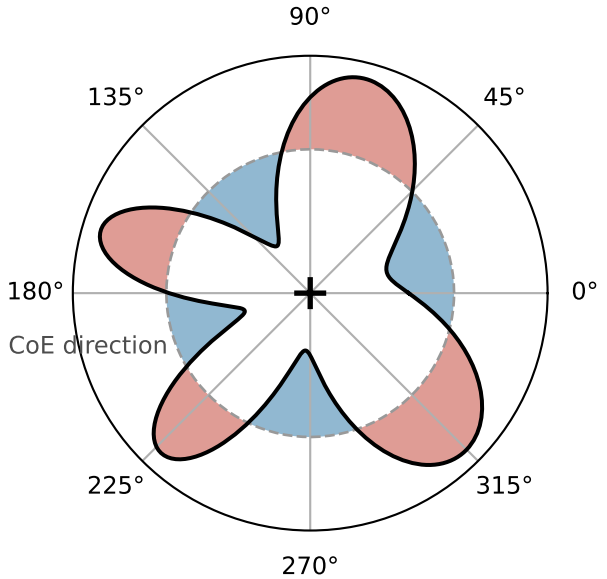
1. The leftover pattern around the ring before any bodies are added (rms 237×)
the flat target (±1×) is a green band around 0, too thin to see at this scale



2. The same pattern split into its waves (modes):
m = 3, 4, 5 (red) carry almost all of it

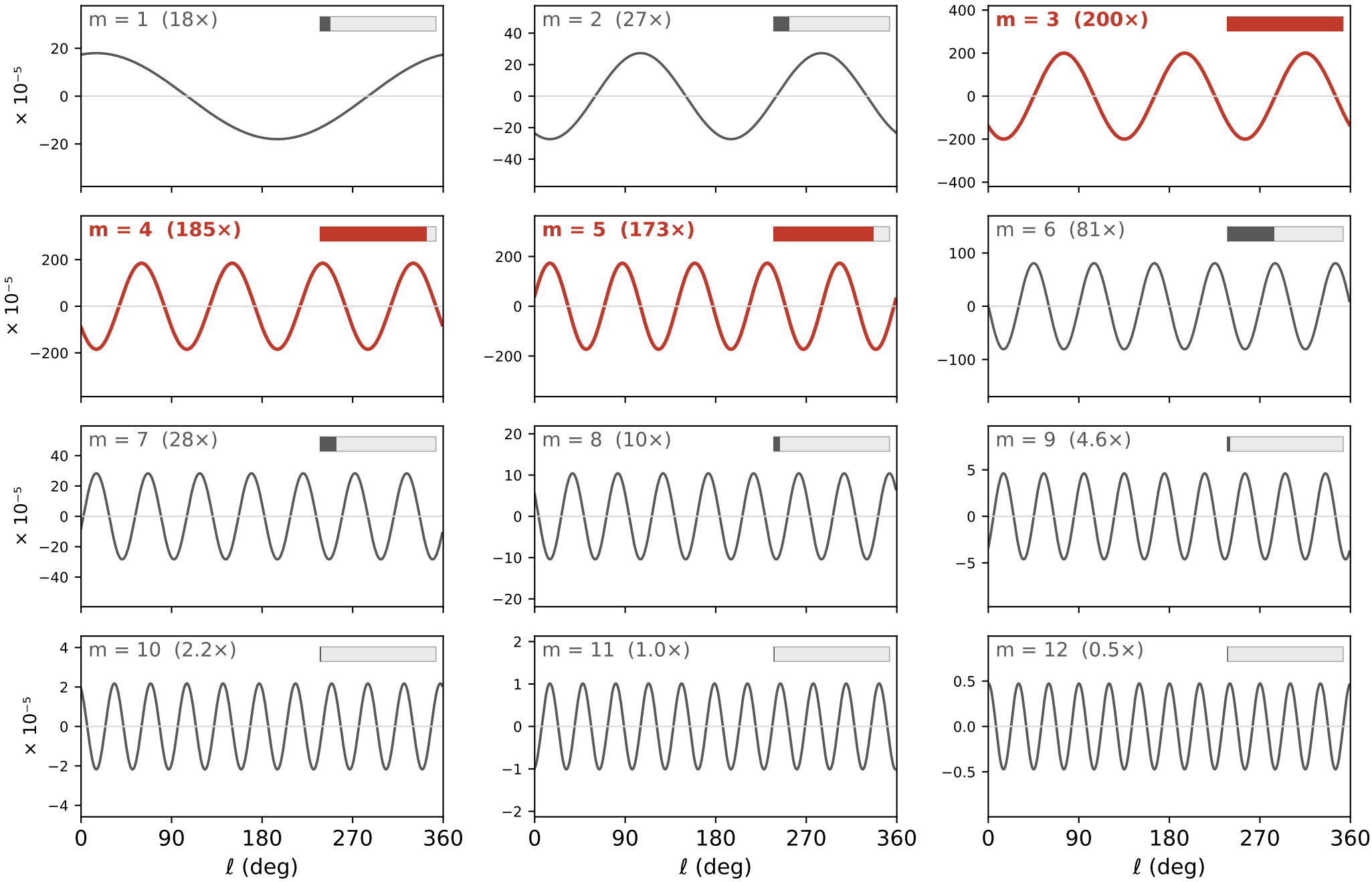


3. The fingerprint drawn around the sky
(dashed = flat; red = above, blue = below)

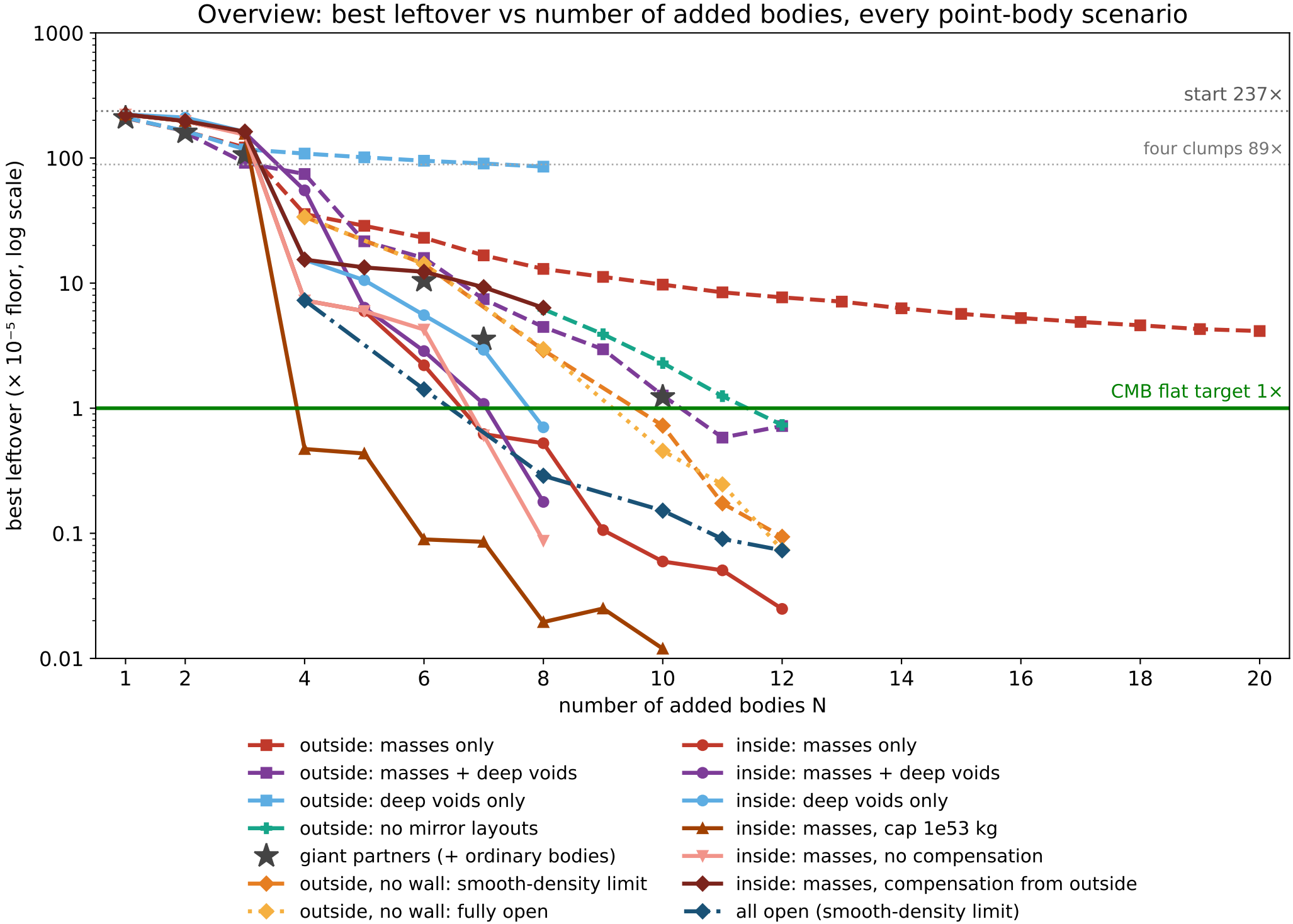


This leftover pattern is the fingerprint. Every scenario in this report is an attempt to cancel it by placing masses and voids.
1× = flat: the pattern squeezed down to the green band.

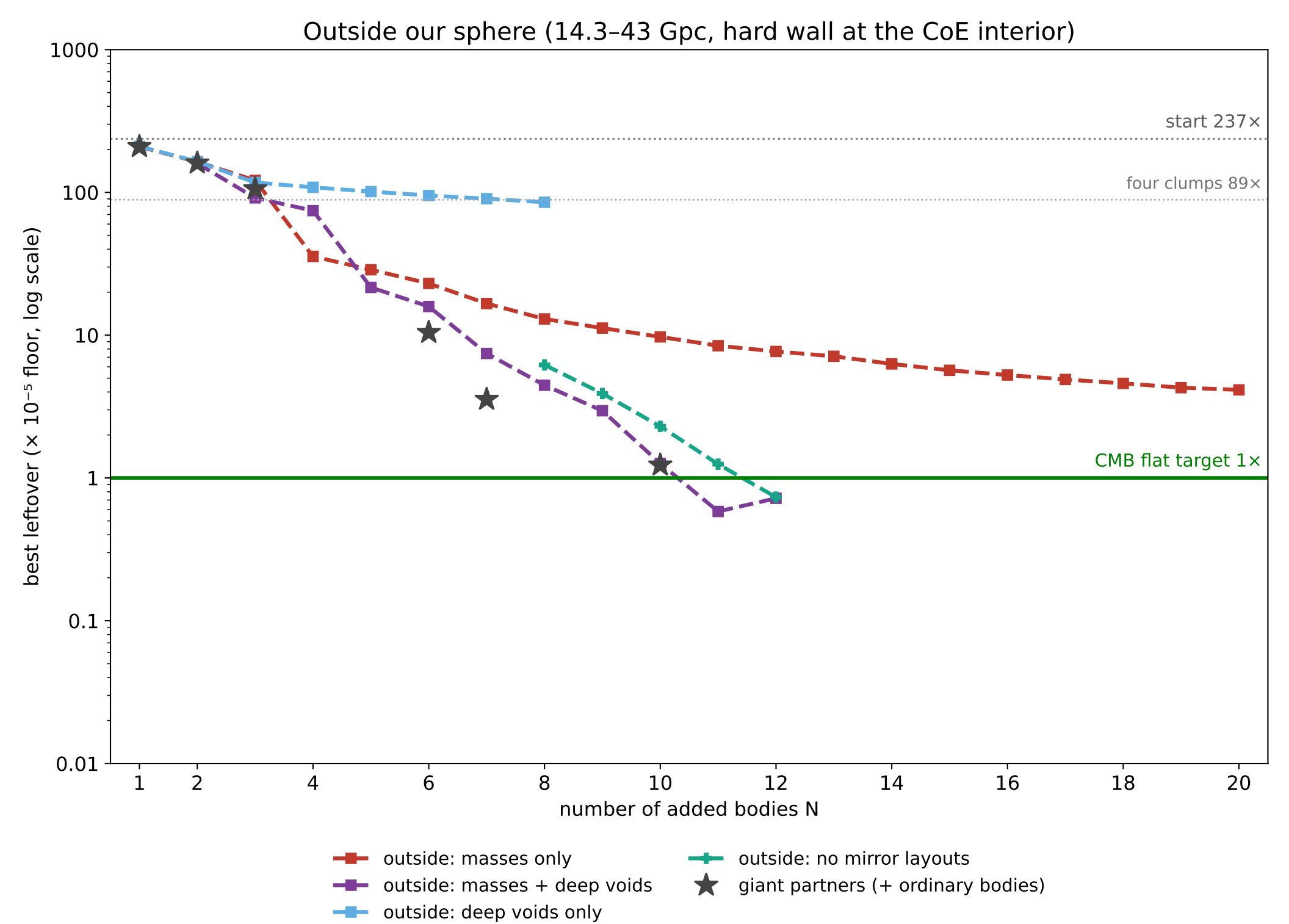
The fingerprint, wave by wave (m = 1…12)



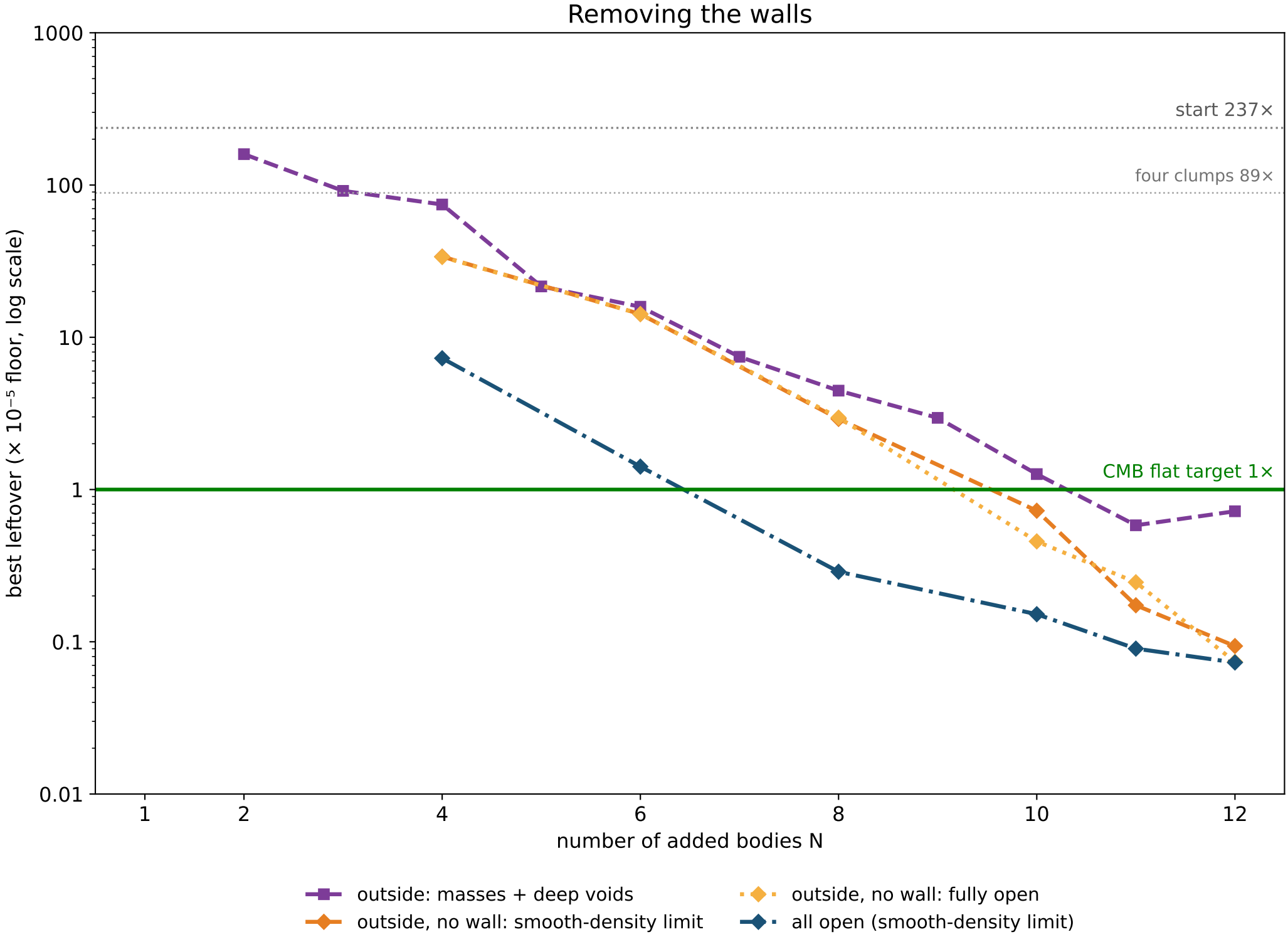
Each panel has its own scale; the numbers give the size, and the bar at top right shows it relative to m = 3 (full bar = 200×).
Adding these 12 waves back together gives the curve on the previous page. m = 3, 4, 5 (red) are the big ones.



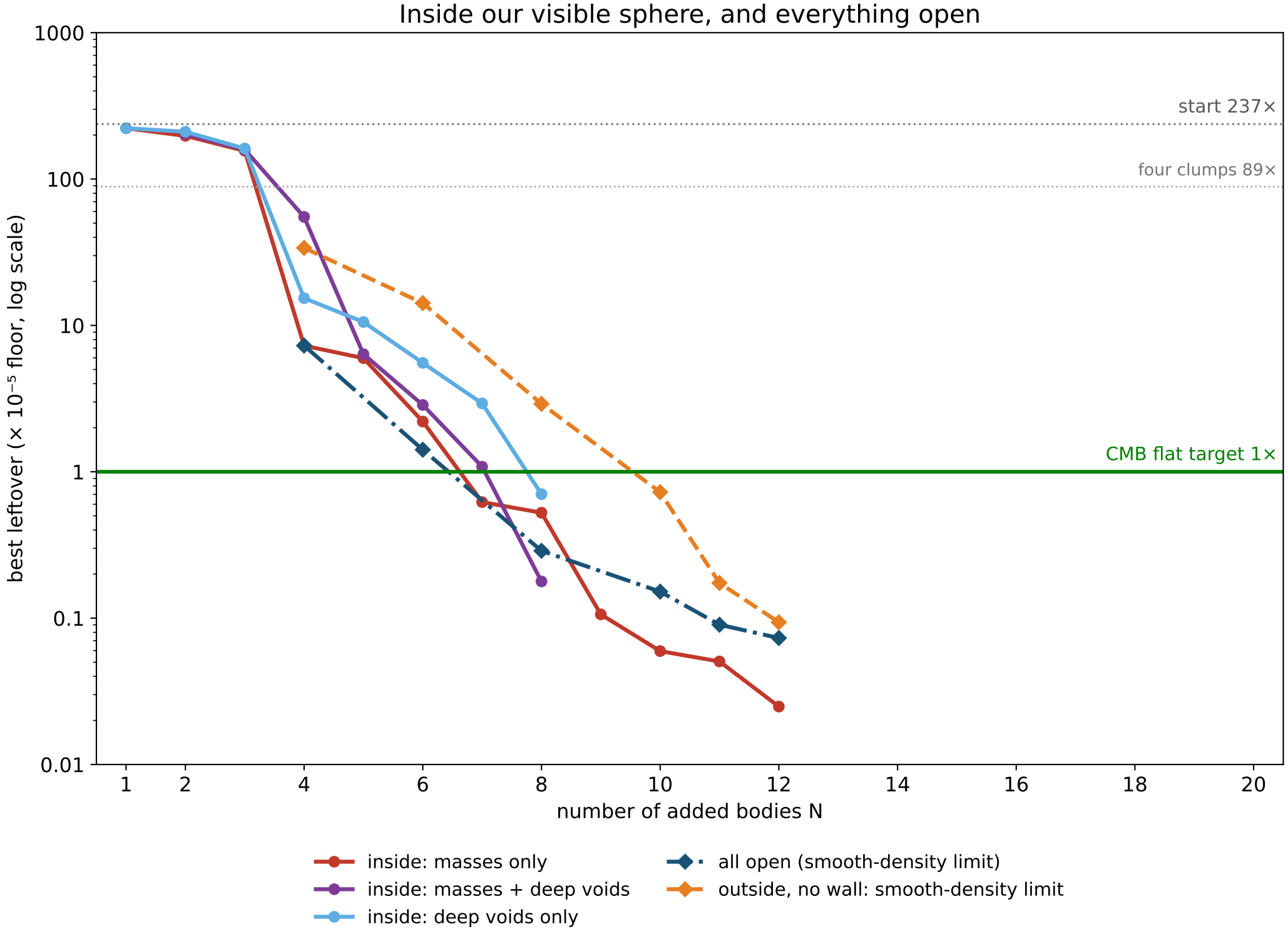
Solid = inside our sphere, dashed = outside, dash-dot = all open, stars = giant partners. The next pages split these into smaller groups.



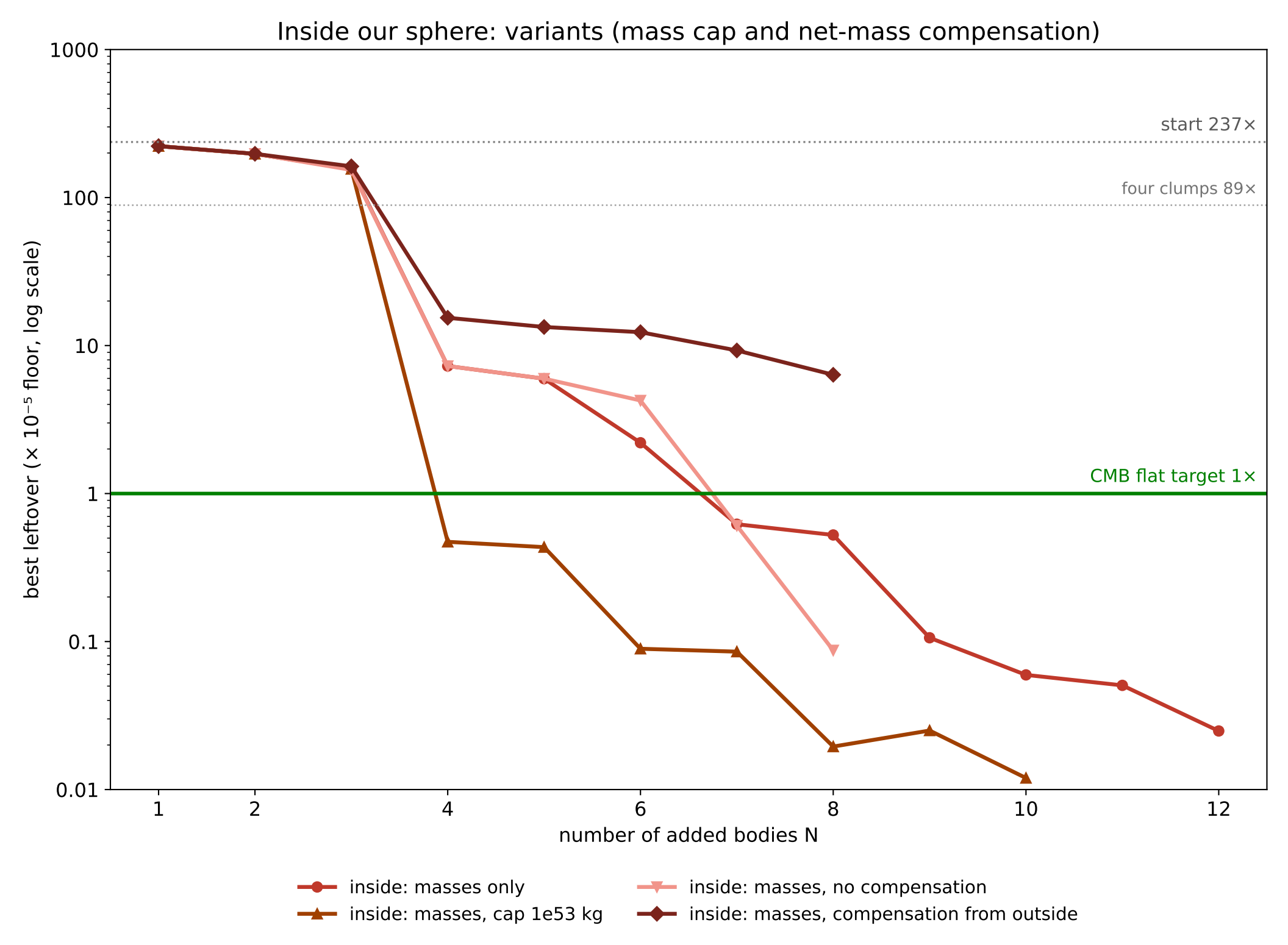
Masses alone stall near 4×; masses + deep voids reach 1× at 11 bodies (12 with mirror layouts forbidden); voids alone stay near 85×. Stars: giant partners (N = partners + ordinary bodies).



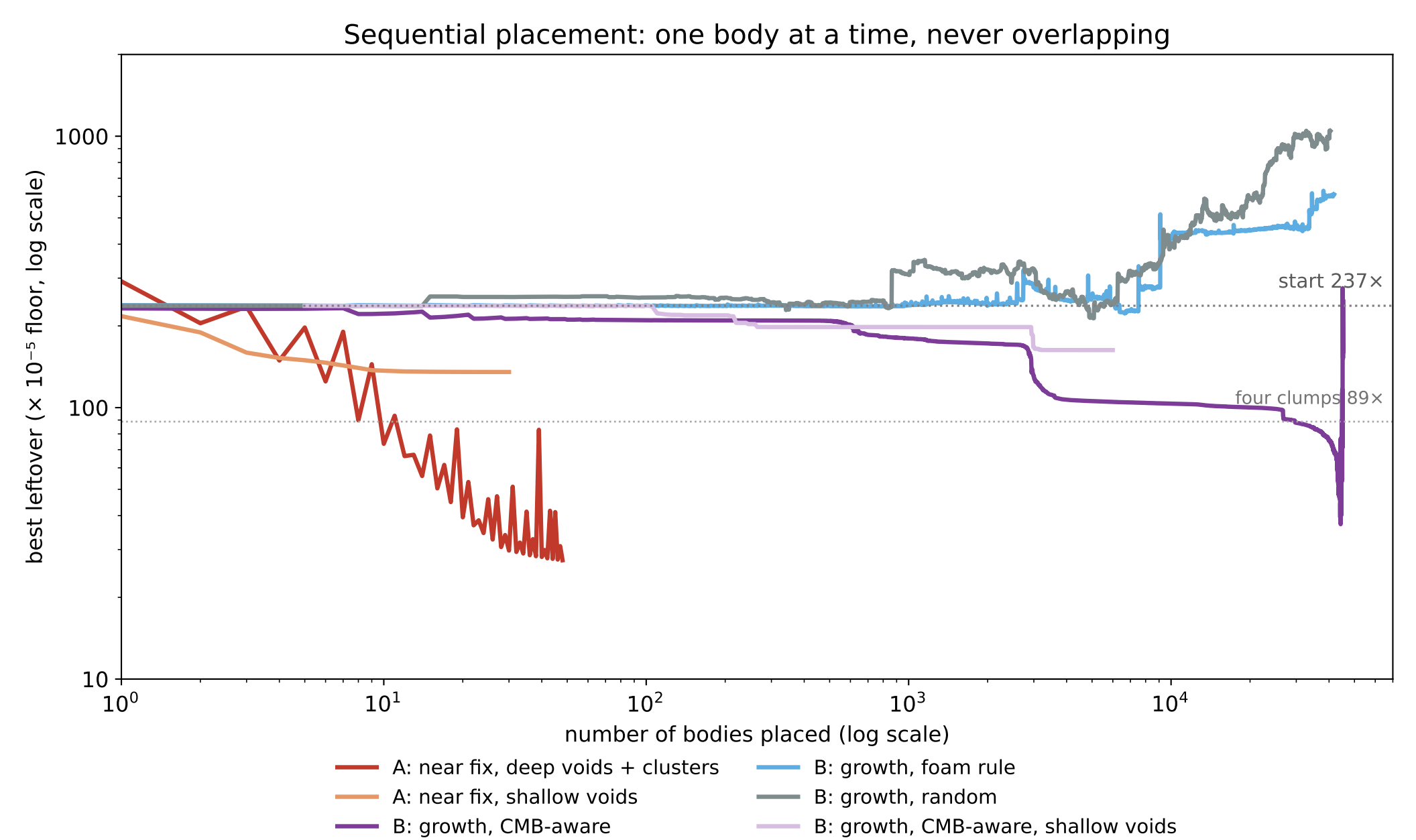
Removing the CoE-interior wall gets to 1× one body sooner (N = 10). Opening our visible sphere too ('all open') gets there at N = 8 — the bodies move inside our sphere.



Inside our sphere: 7 masses reach 1× (0.62×), 4 masses + 4 voids reach 0.18×. Caveat: for bodies between us and the CMB the model leaves out ISW/lensing effects.

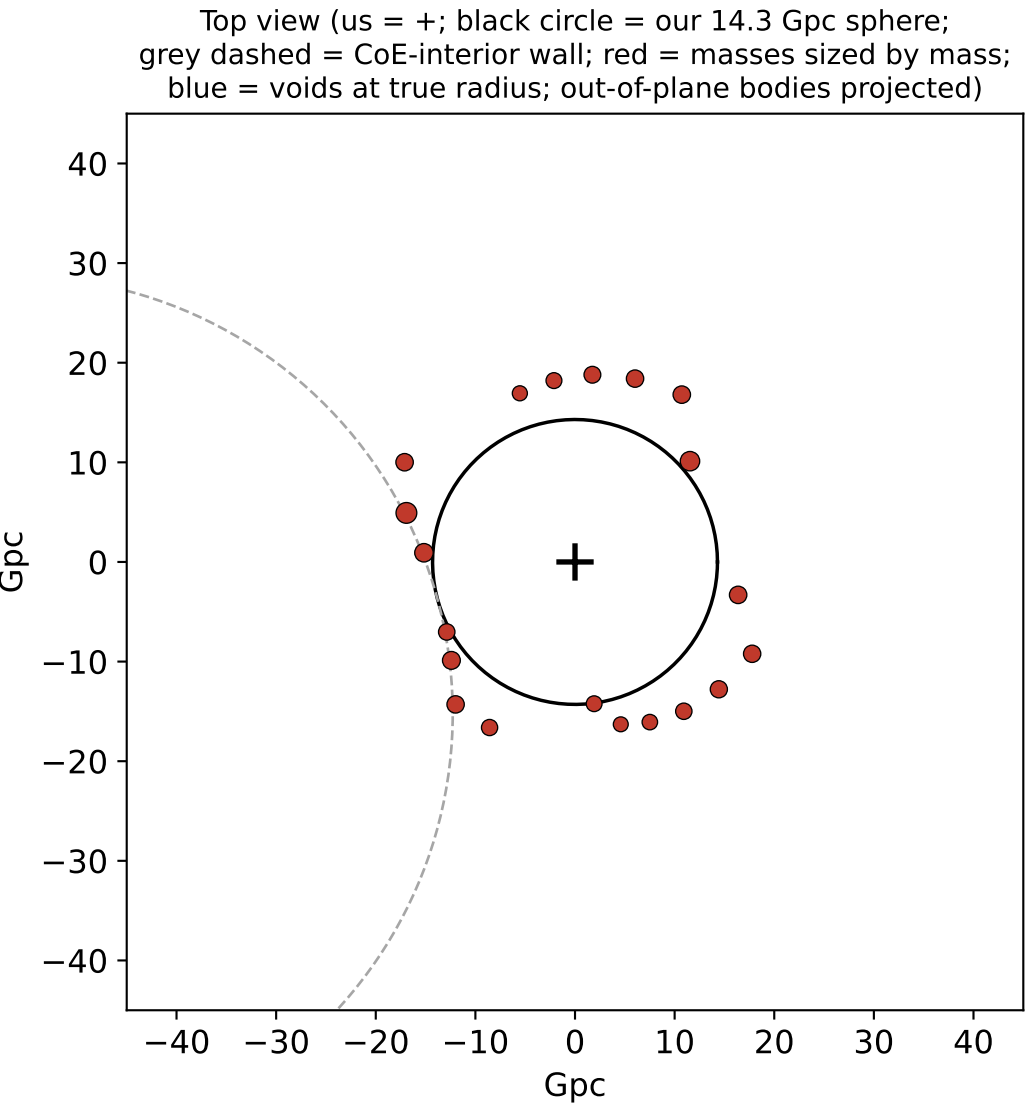


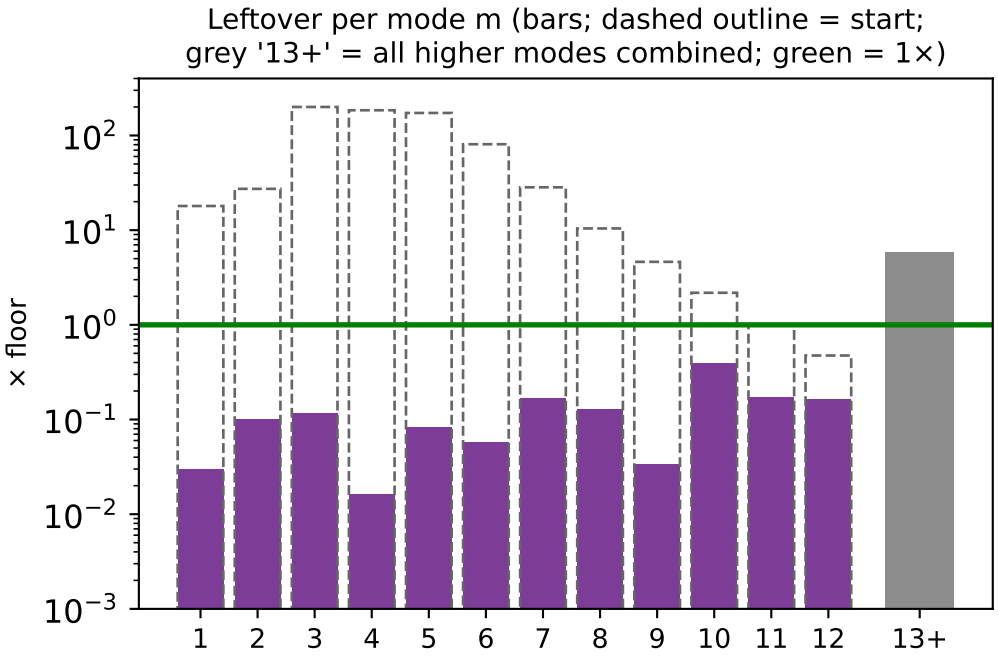
Raising the mass cap to 1e53 kg reaches 1× with only 4 masses; dropping compensation changes little; taking the compensating mass from outside our sphere stalls near 6×.

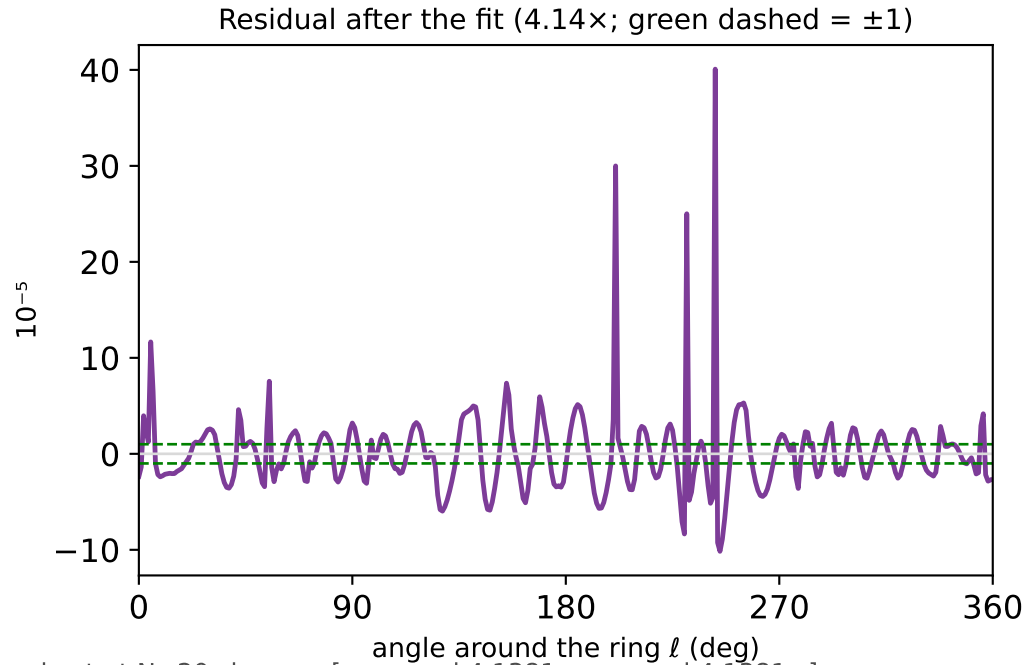


A: the near region (14.3–30 Gpc) fixed pair by pair (best seed). B: growth from empty until the region jams (best seed per rule; coarse 1.5 Gpc voids). None reaches 1×.

Outside: masses only
Example: N = 20 (20 masses), leftover 4.14×

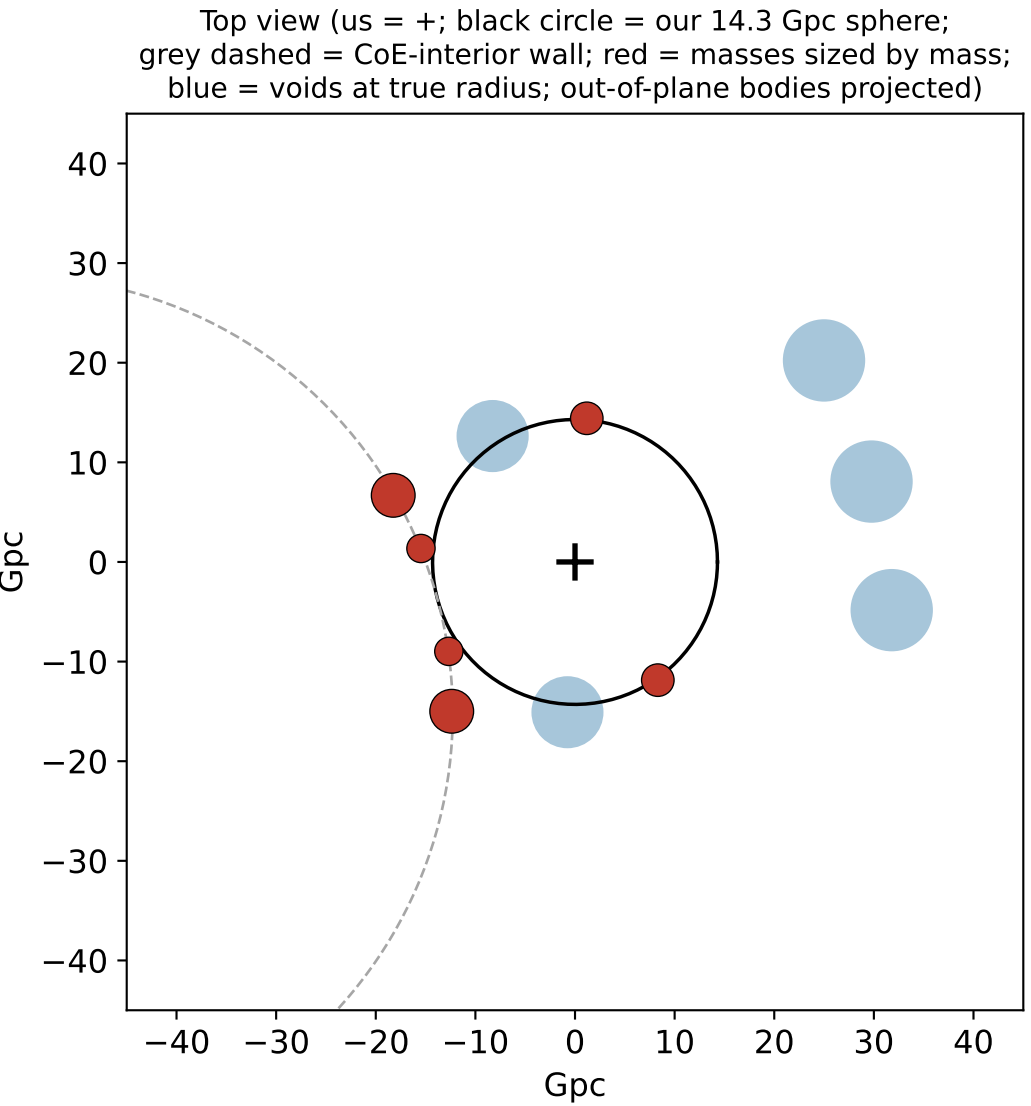


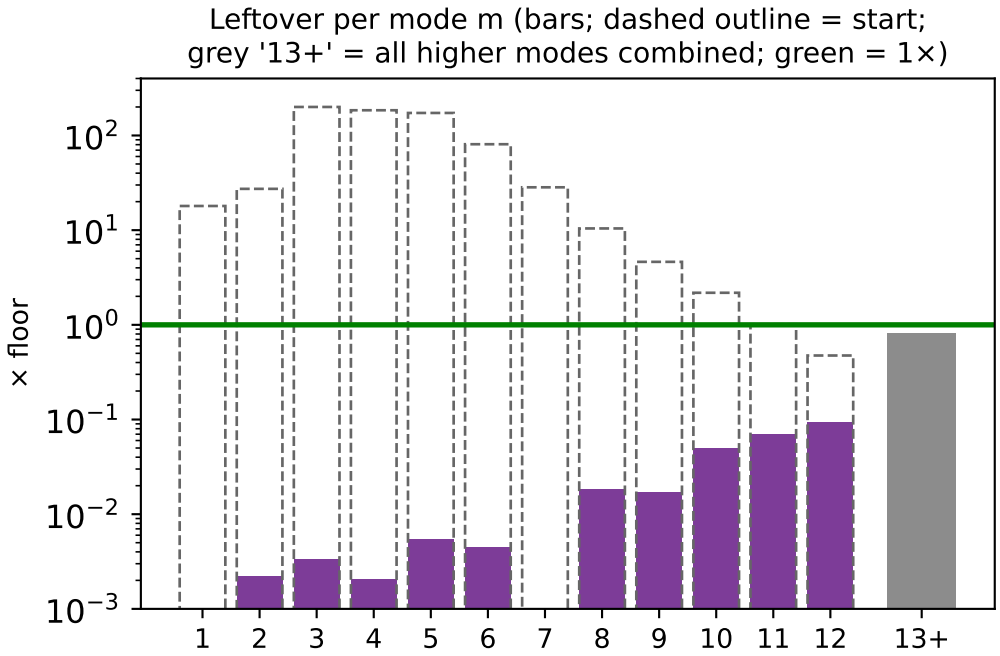


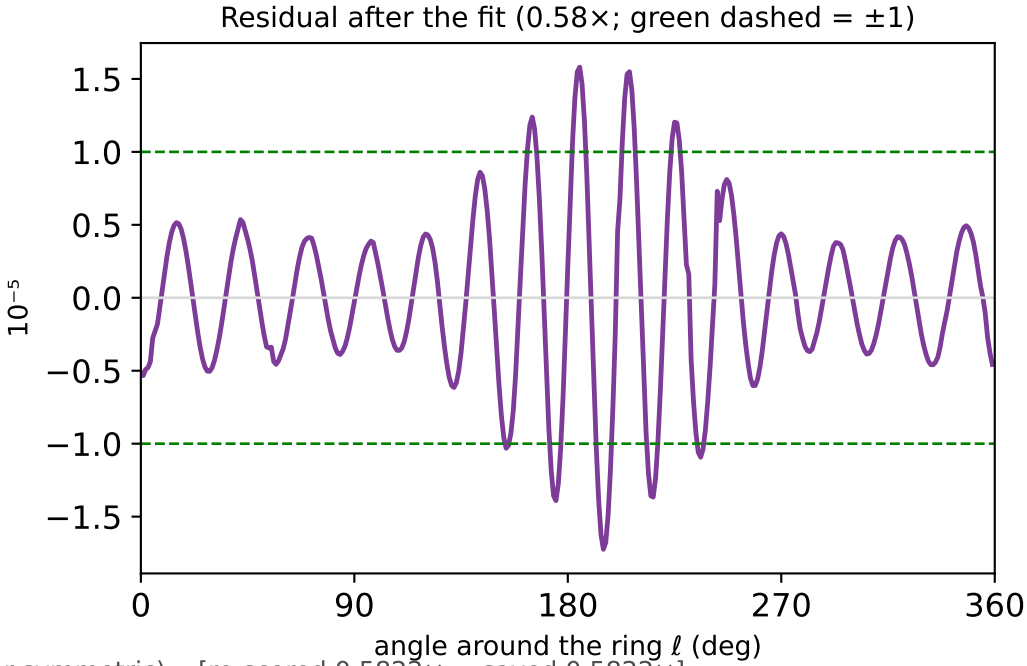


Hard wall at the CoE interior; 14.3–43 Gpc from us. Never reaches 1×; best at N=20 shown. [re-scored 4.1381× = saved 4.1381×]

Outside: masses + deep voids (mirror layouts allowed)
Example: N = 11 (6 masses + 5 voids), leftover 0.58×

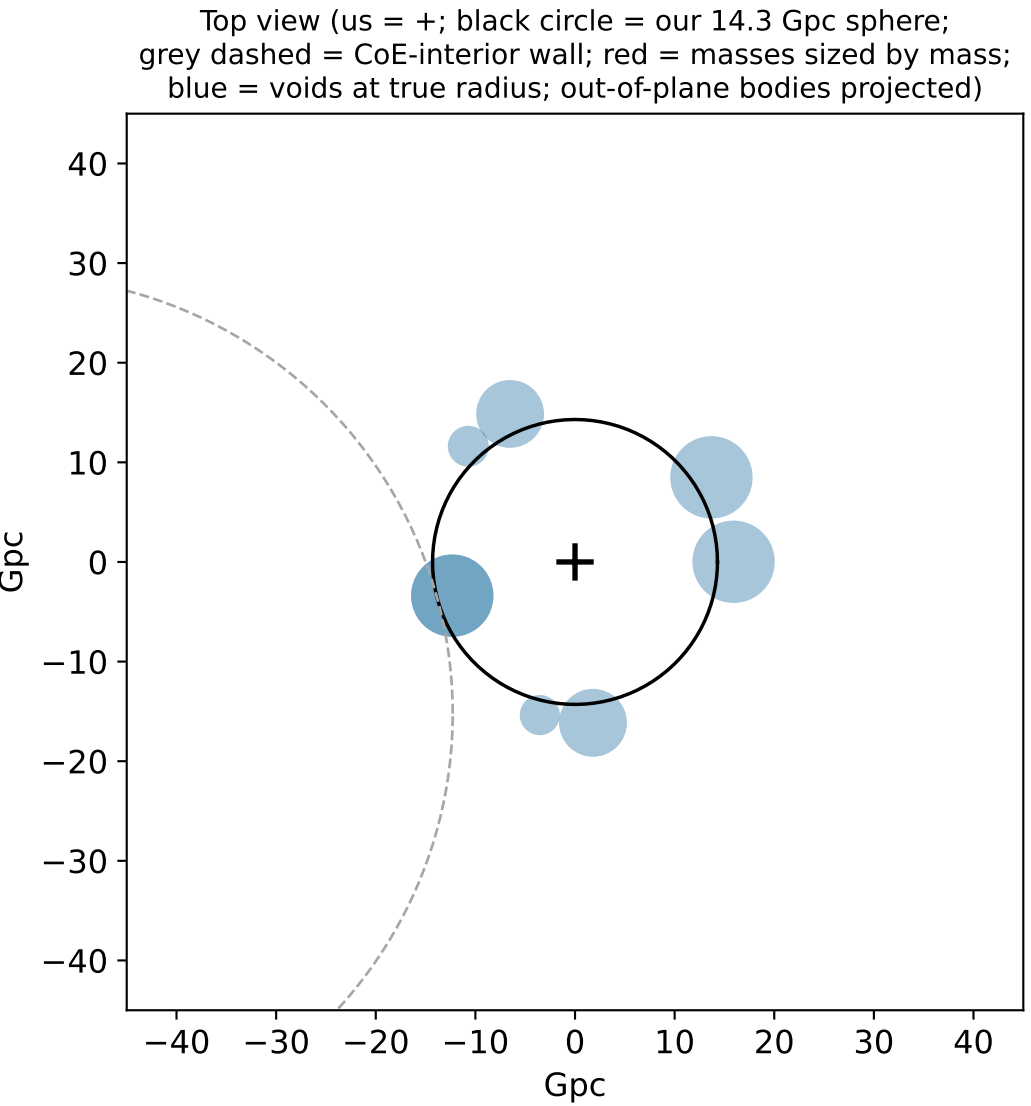


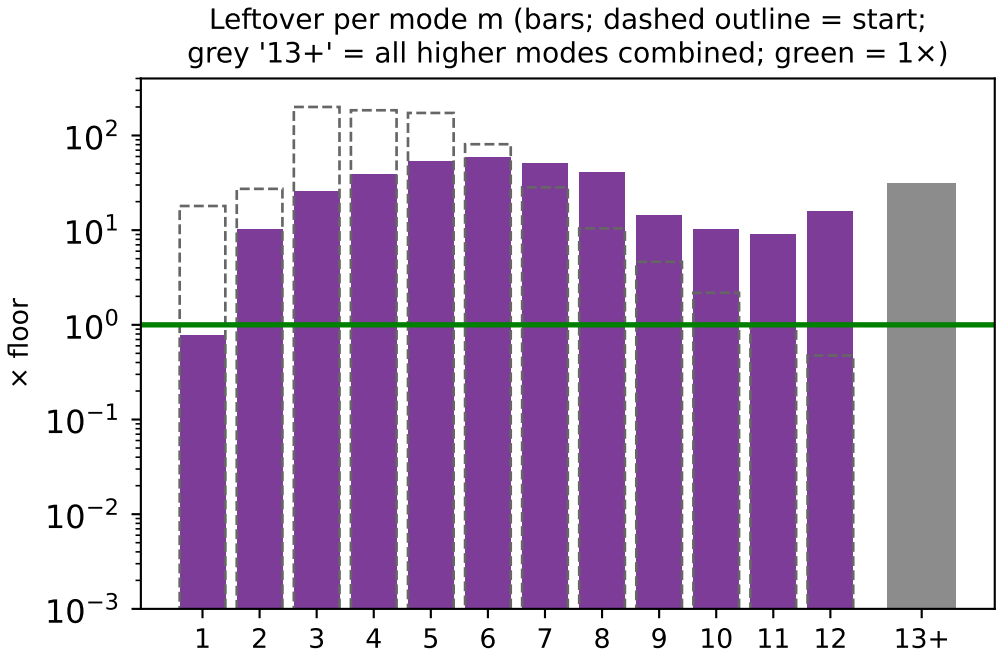




Hard wall; first below 1× at N=11 (layout came out mirror-symmetric). [re-scored 0.5822× = saved 0.5822×]

Outside: deep voids only
Example: N = 8 (8 voids), leftover 85.30×

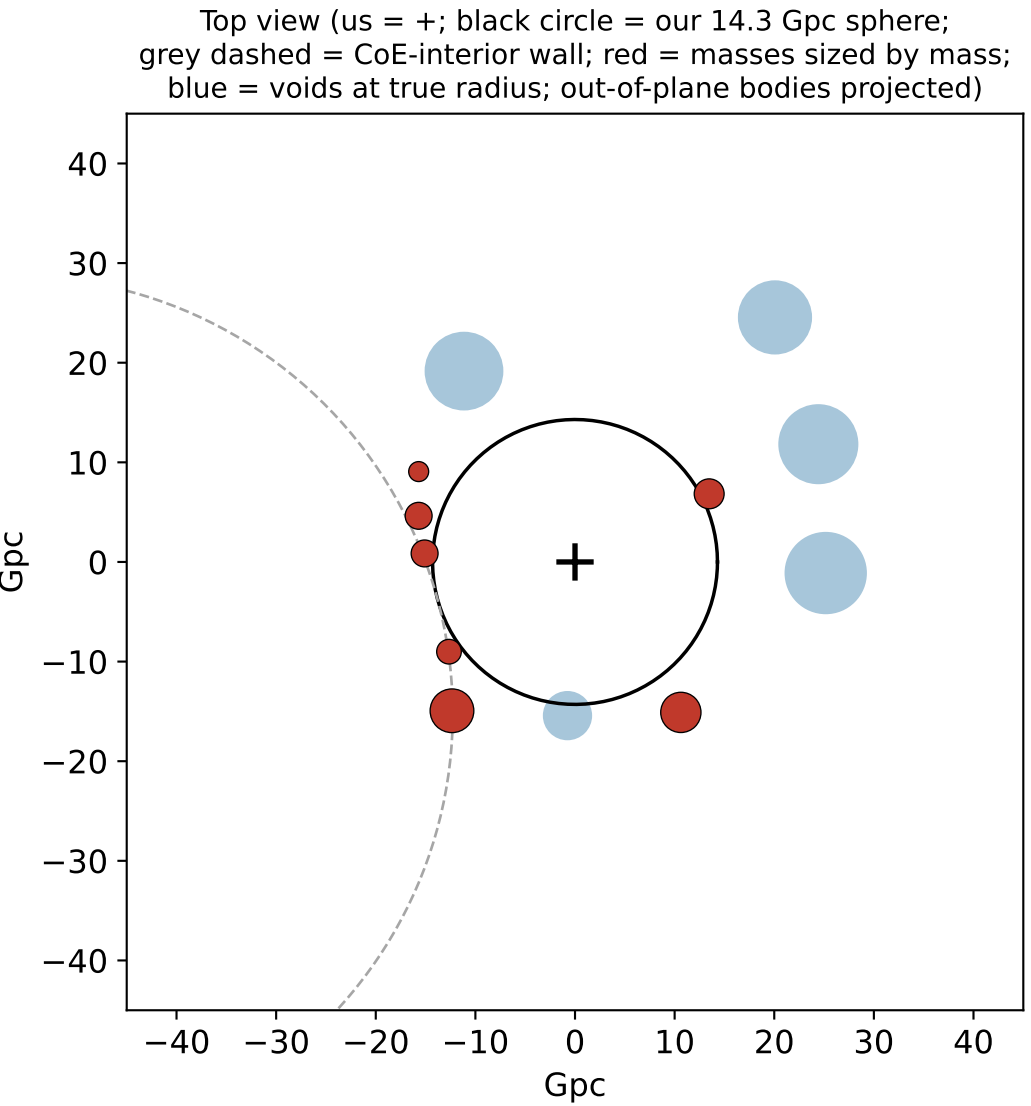


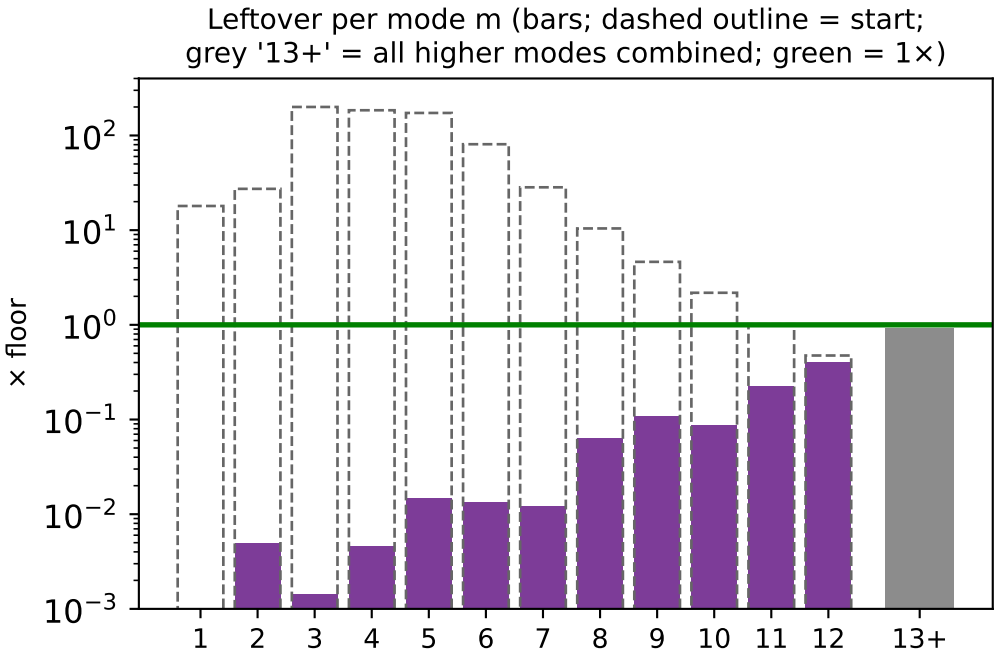


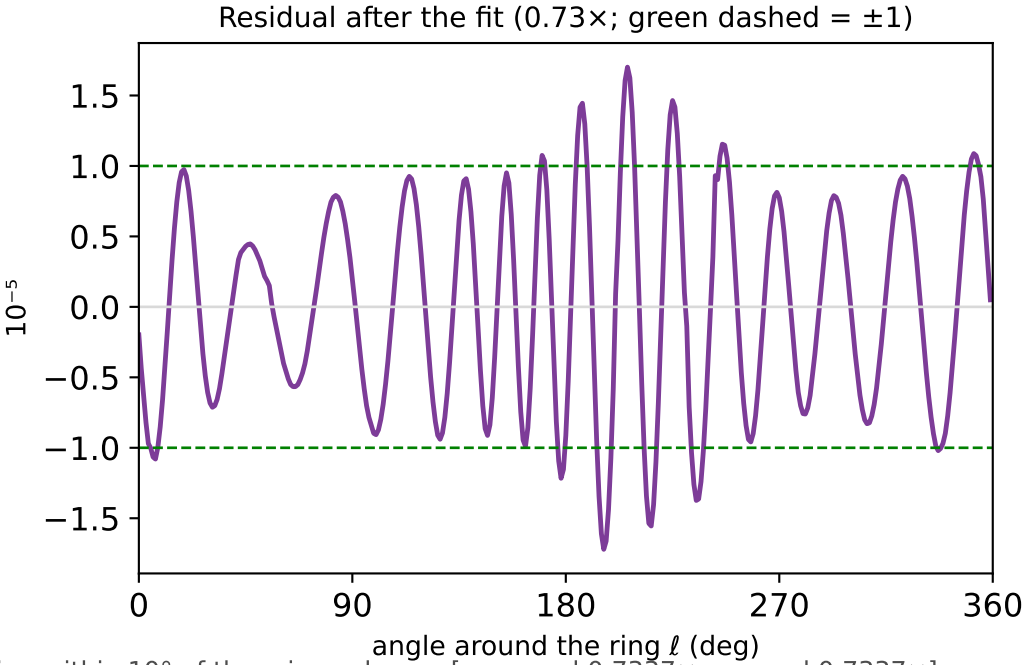


Voids capped at 4 Gpc; never beats the four clumps. [re-scored 85.3046× = saved 85.3046×]

Outside: masses + deep voids, no mirror layouts
Example: N = 12 (7 masses + 5 voids), leftover 0.73×



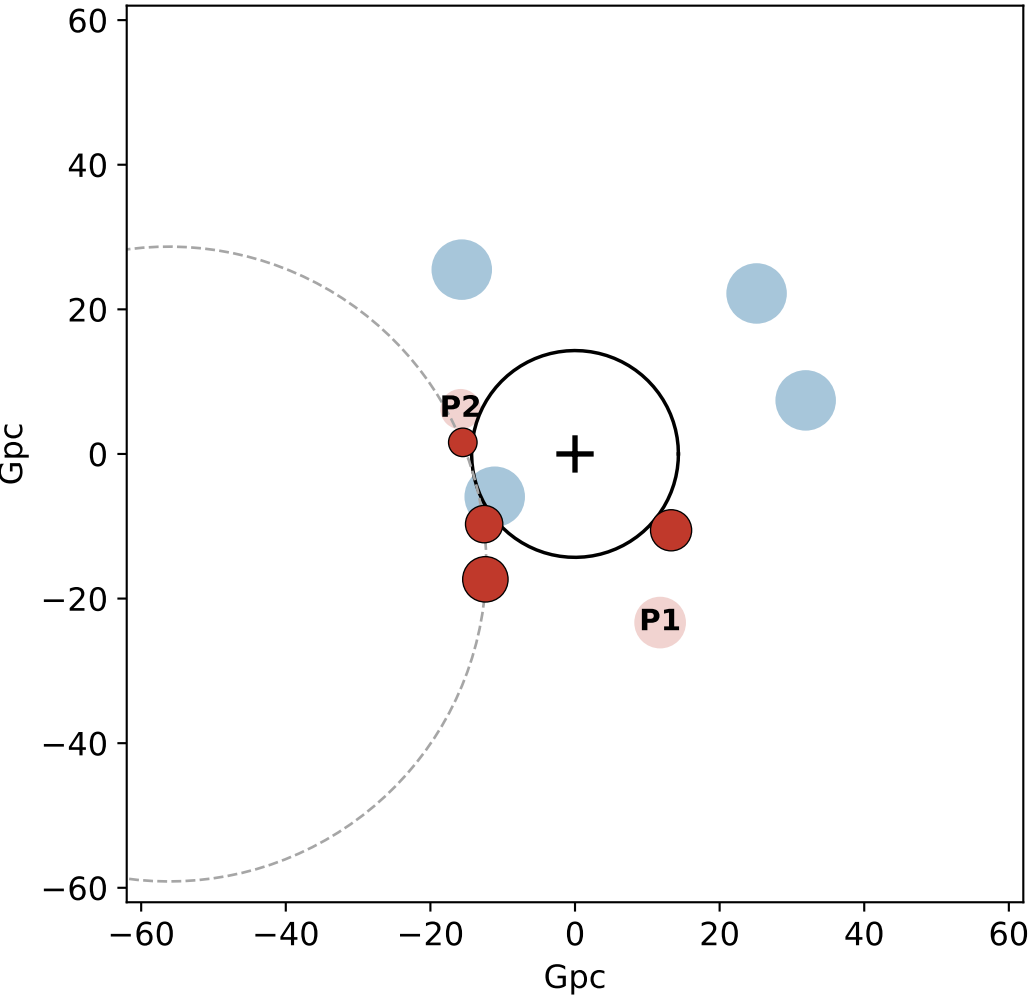




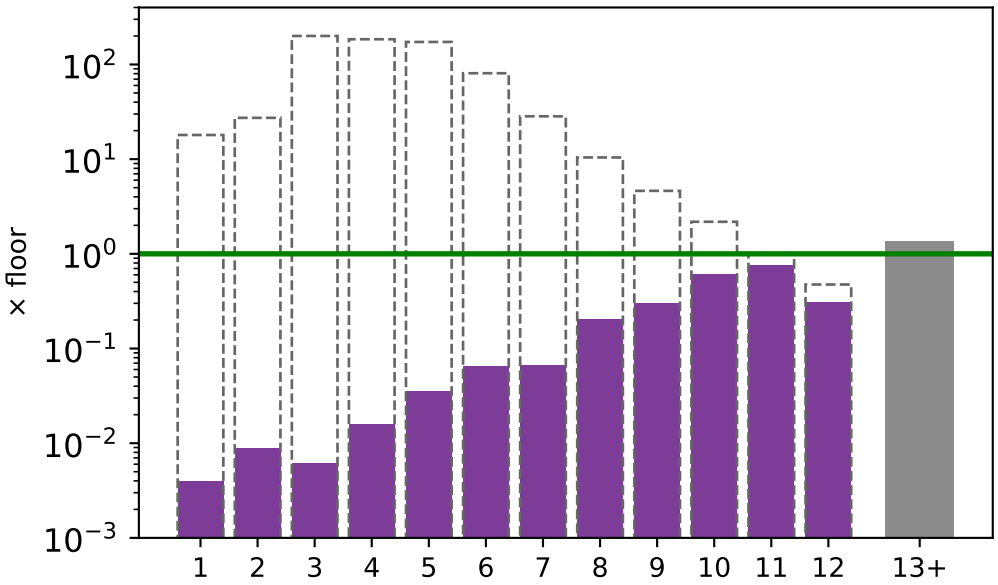
No same-kind pair within 20° / 15% / 25% of each other's mirror image; nothing within 10° of the mirror plane. [re-scored 0.7337× = saved 0.7337×]

Giant partners (our sphere over/underdense + 1–3 giant partners, ± ordinary bodies)
Example: N = 10 (4 masses + 4 voids, 2 giant partners), leftover 1.23×

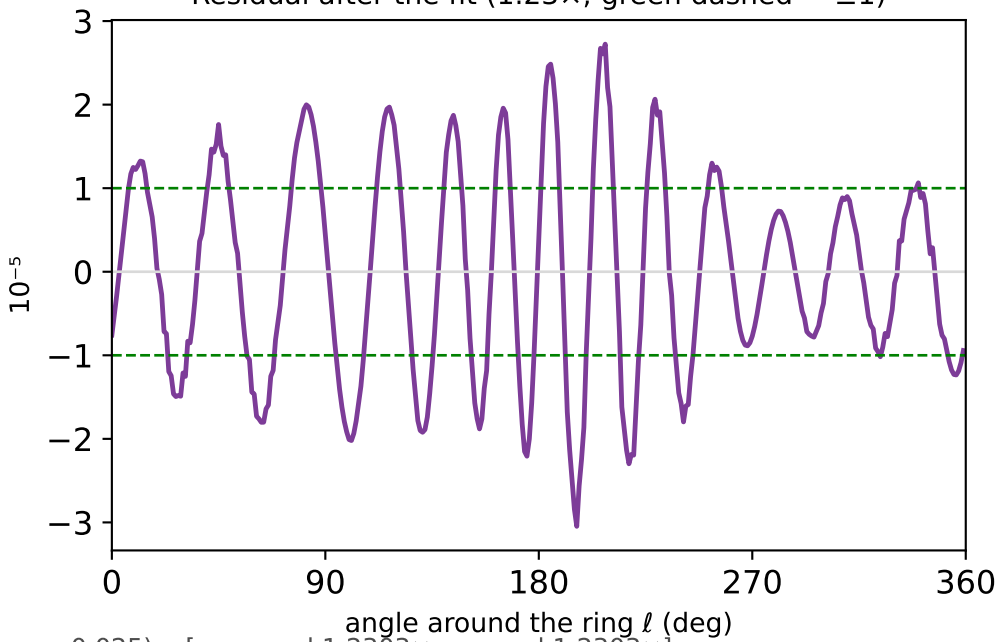
Top view (us = +; black circle = our 14.3 Gpc sphere;
grey dashed = CoE-interior wall; red = masses sized by mass;
blue = voids at true radius; out-of-plane bodies projected)



Leftover per mode m (bars; dashed outline = start;
grey '13+' = all higher modes combined; green = 1×)

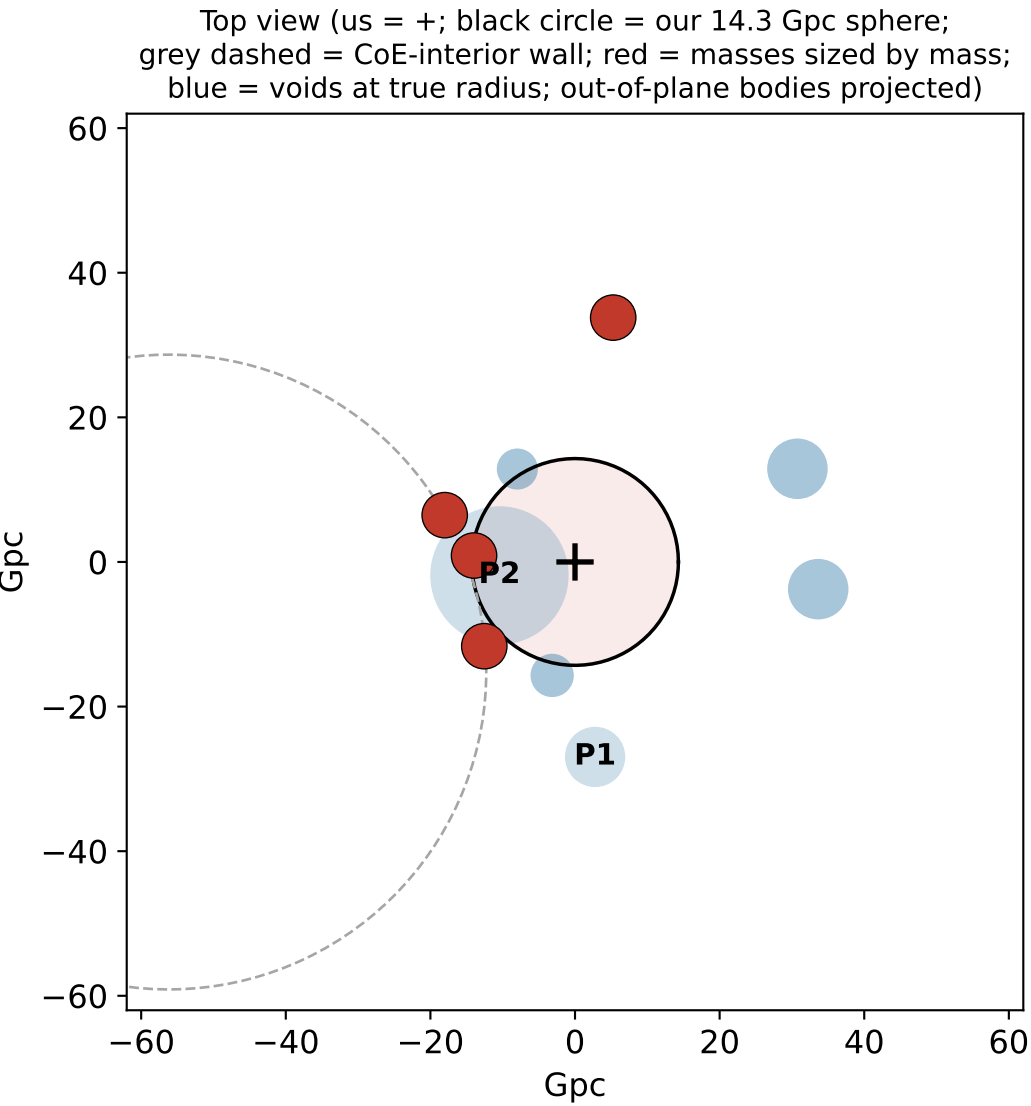


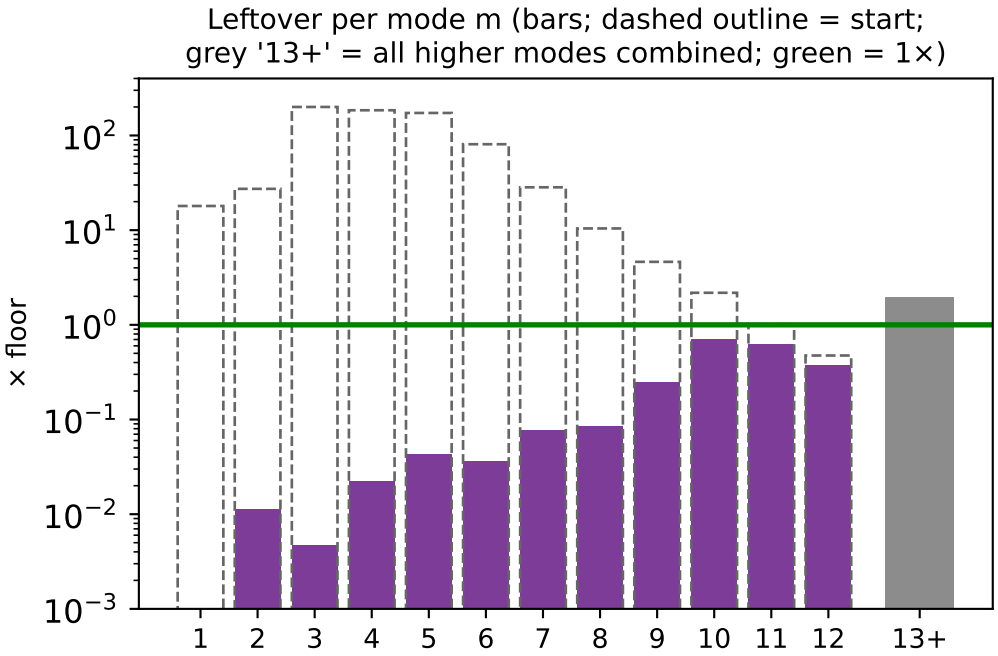
Residual after the fit (1.23×; green dashed = ±1)

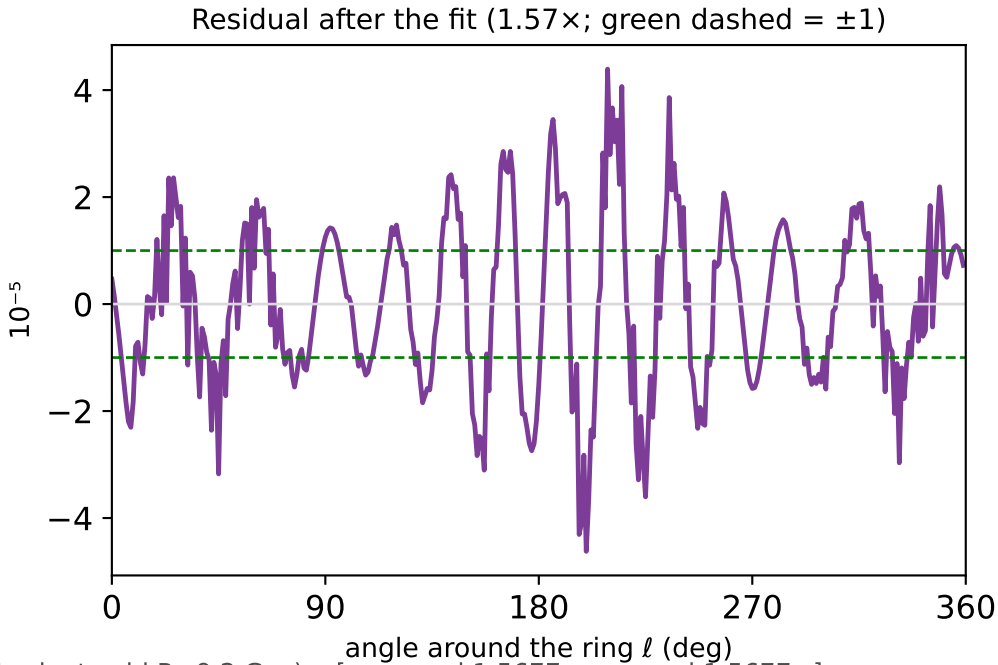


N counts partners + ordinary bodies. Best: free_2p+4C4V (δ_us=-0.025). [re-scored 1.2303× = saved 1.2303×]

Giant partners (our sphere over/underdense + 1–3 giant partners, ± ordinary bodies)
Second example: best layout with a giant partner, leftover 1.57×

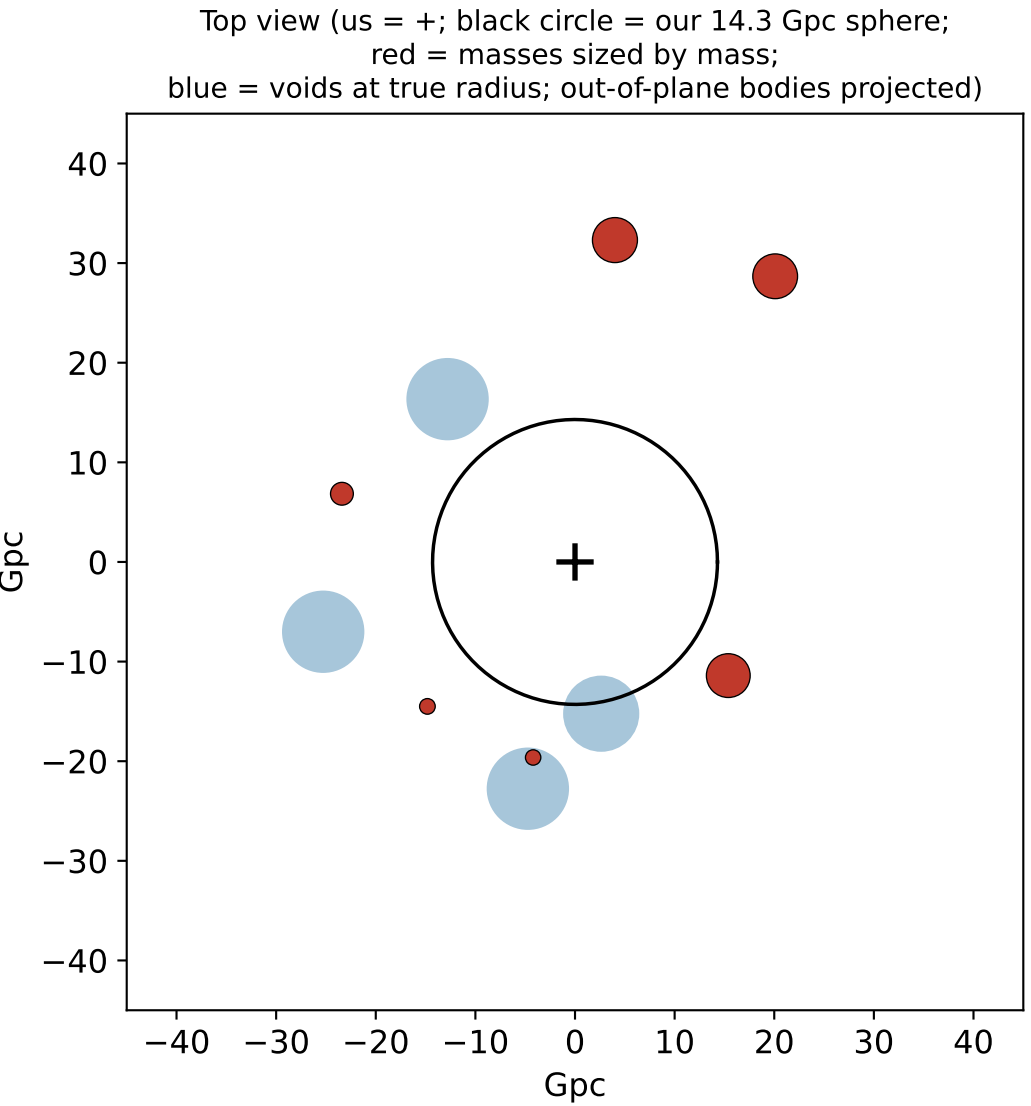


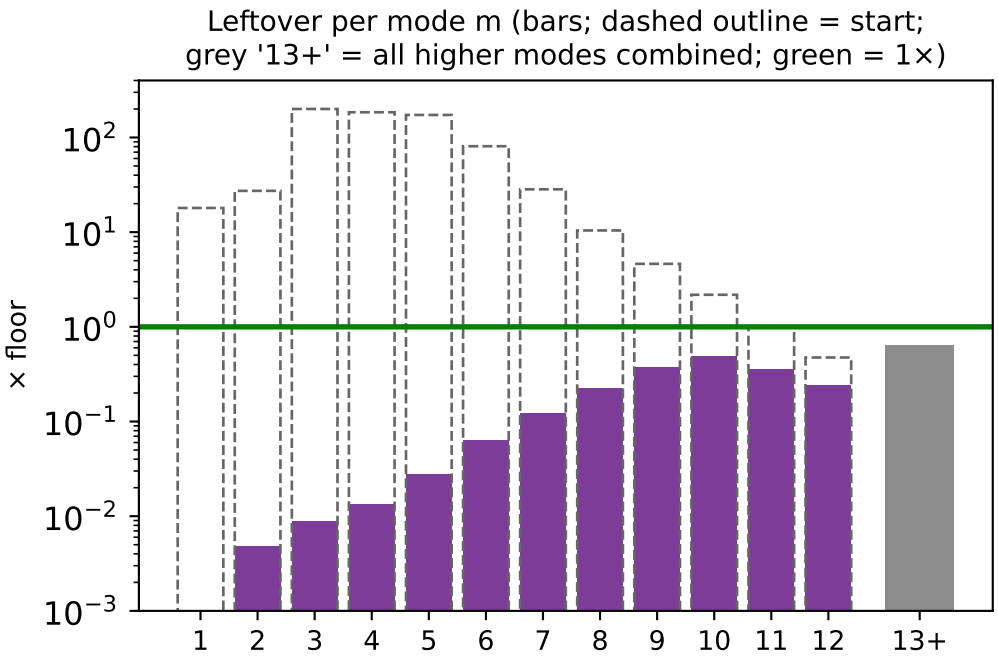


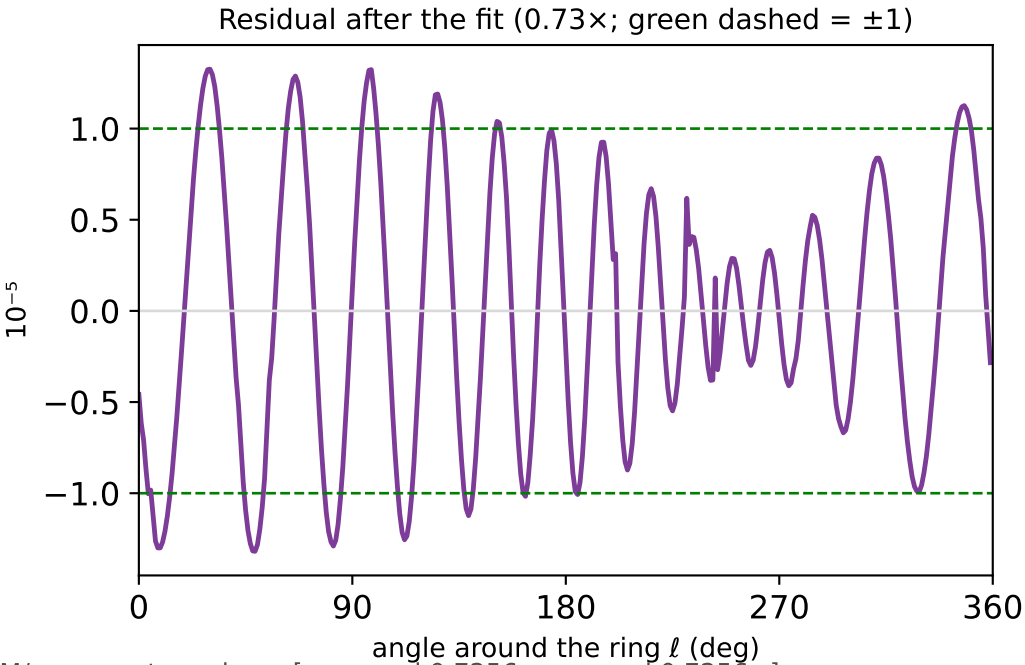


Best layout that really uses a giant partner: over_2p+4C4V (δ_us=+0.21, giant void R=9.2 Gpc). [re-scored 1.5677× = saved 1.5677×]

Outside, no wall: smooth-density limit
Example: N = 10 (6 masses + 4 voids), leftover 0.73×

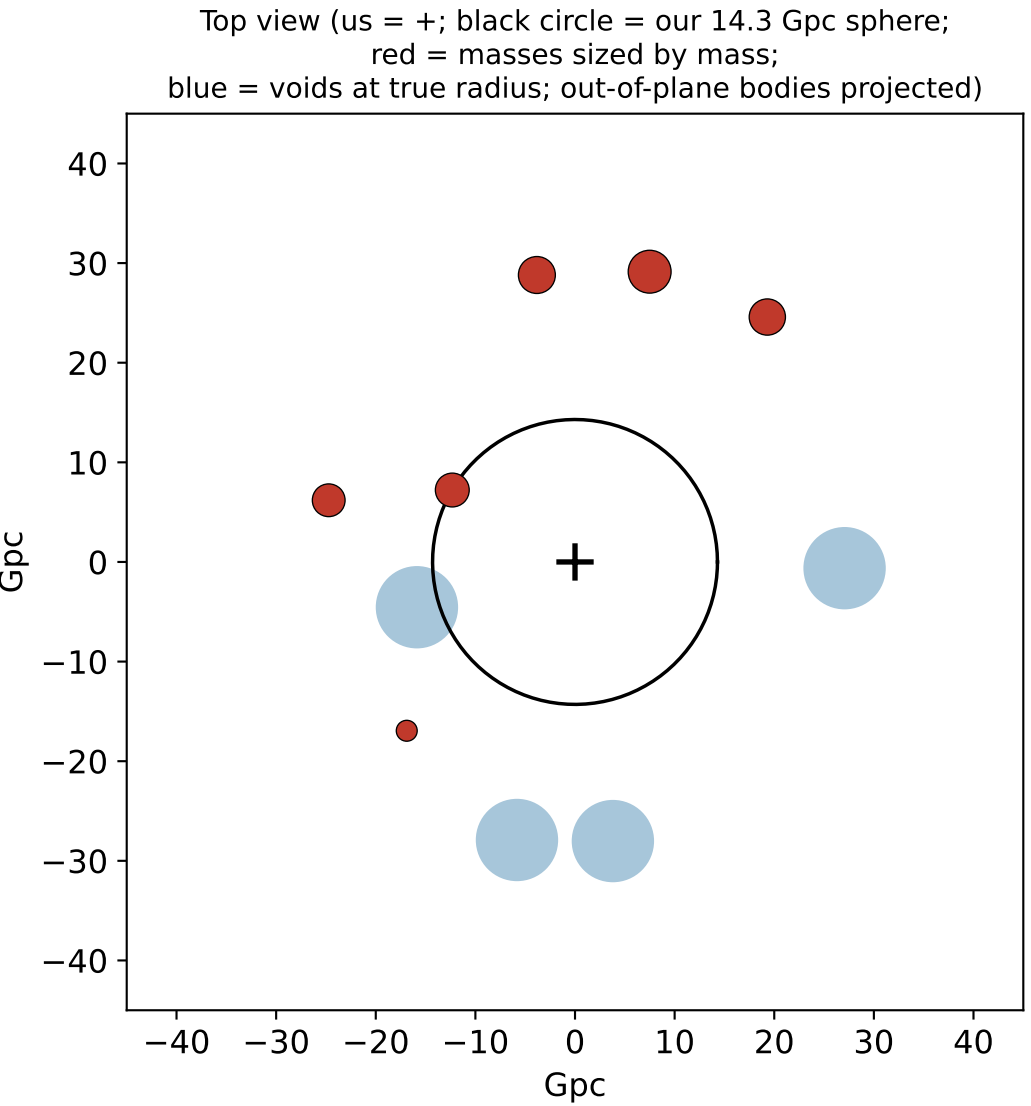


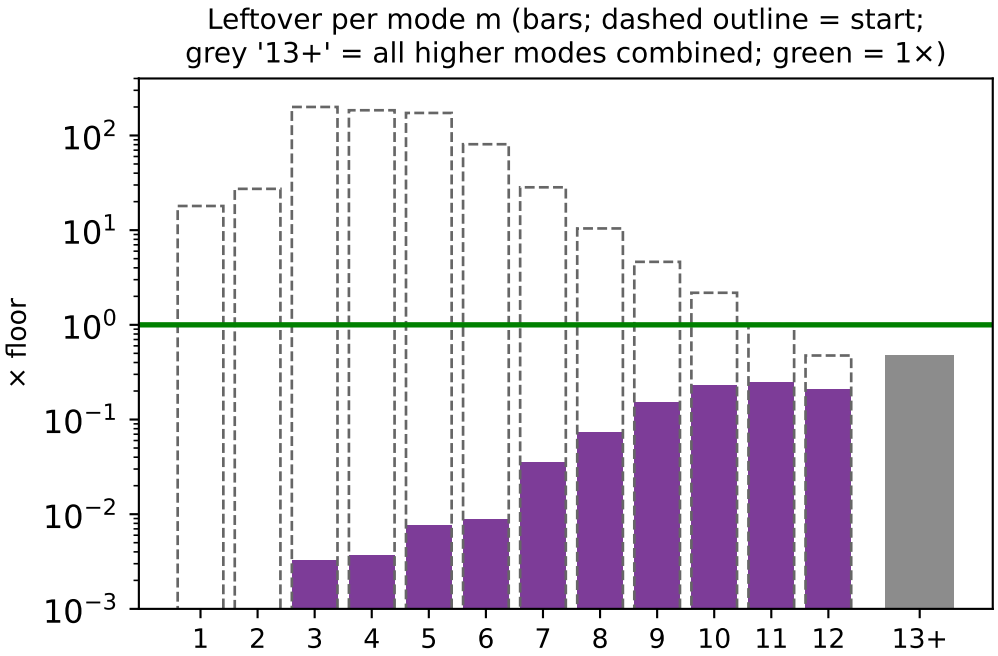


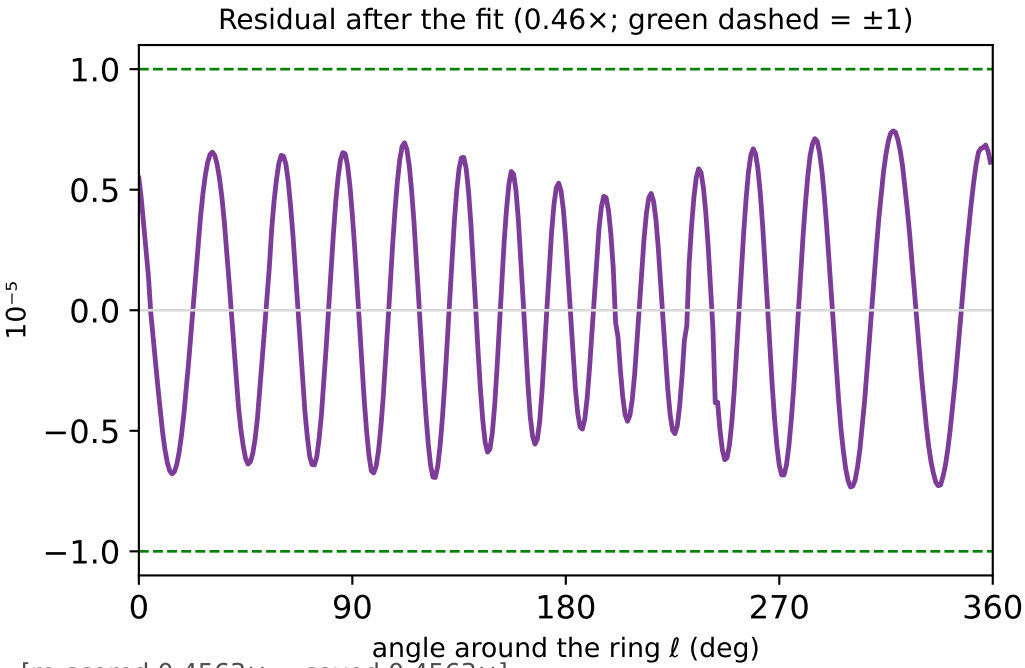


CoE-interior wall removed; mass ≤ 2e52·ρ(x)/ρ̄, supply volumes M/ρ may not overlap. [re-scored 0.7256× = saved 0.7256×]

Outside, no wall: fully open
Example: N = 10 (6 masses + 4 voids), leftover 0.46×

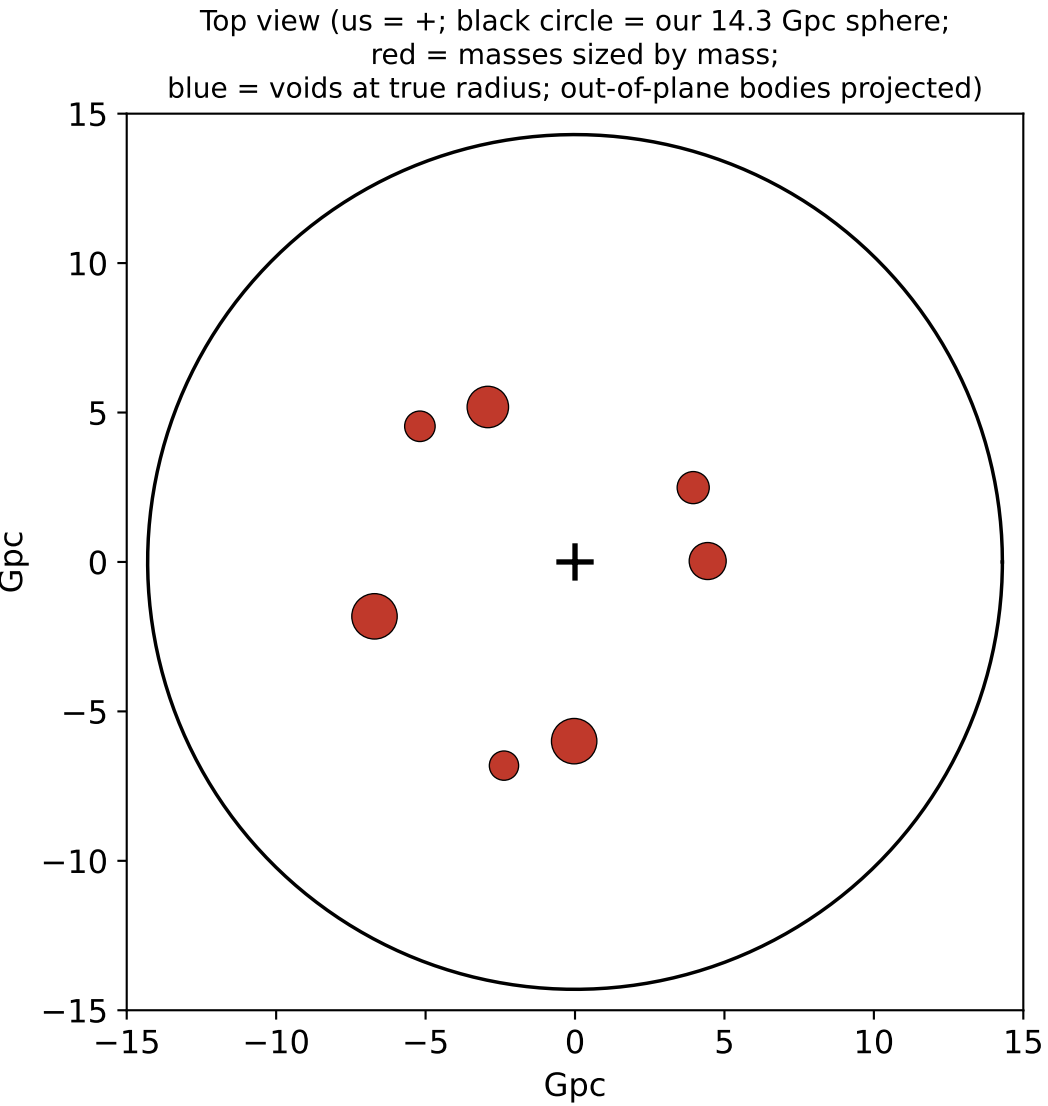


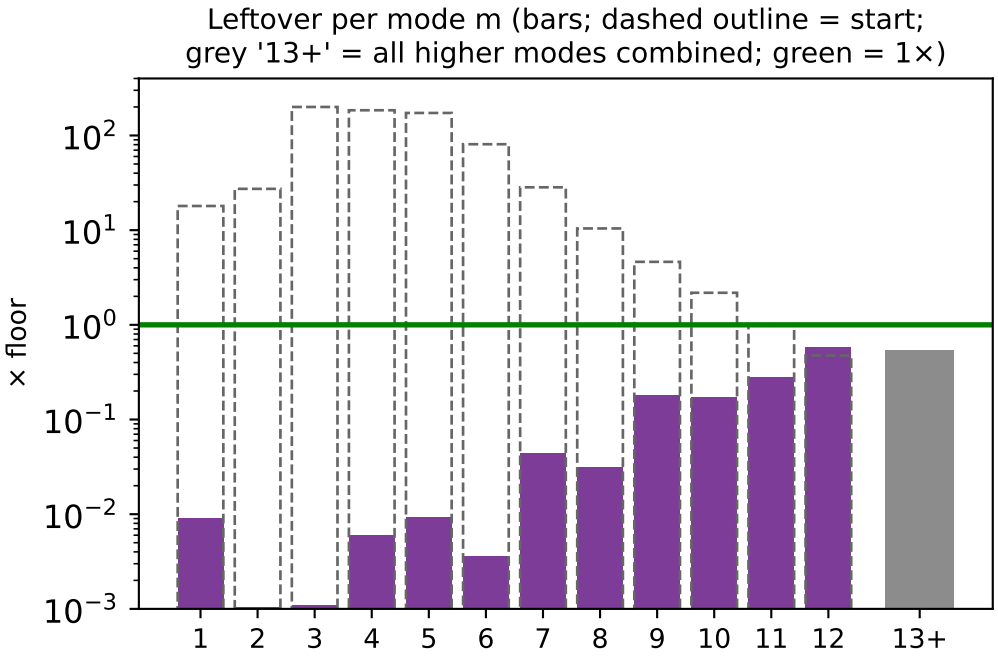


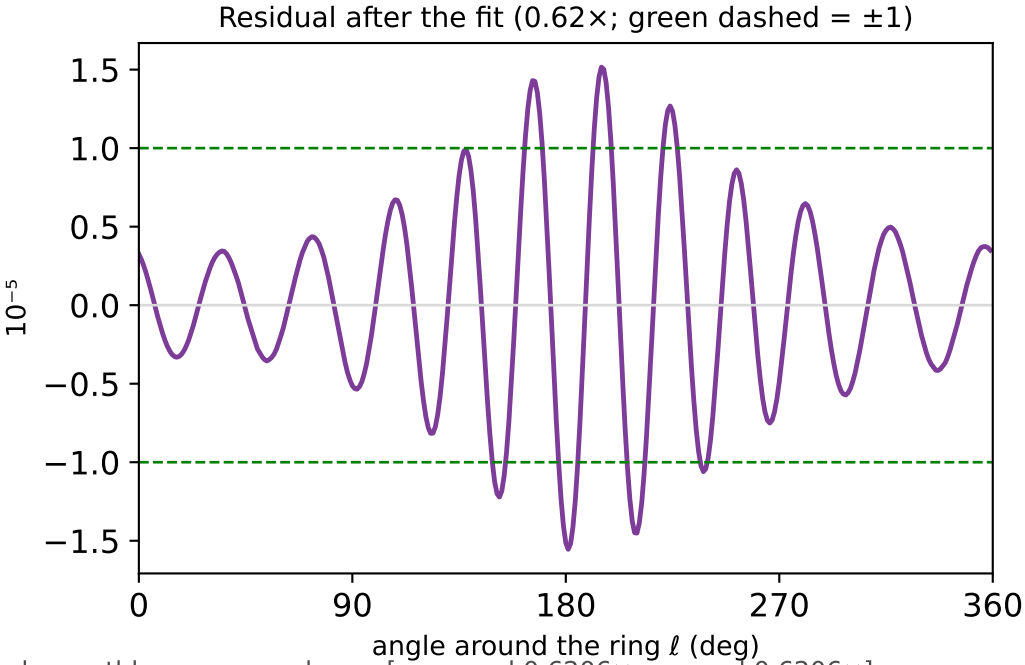


CoE-interior wall removed; usual caps only. [re-scored 0.4563× = saved 0.4563×]

Inside our sphere: masses only
Example: N = 7 (7 masses), leftover 0.62×

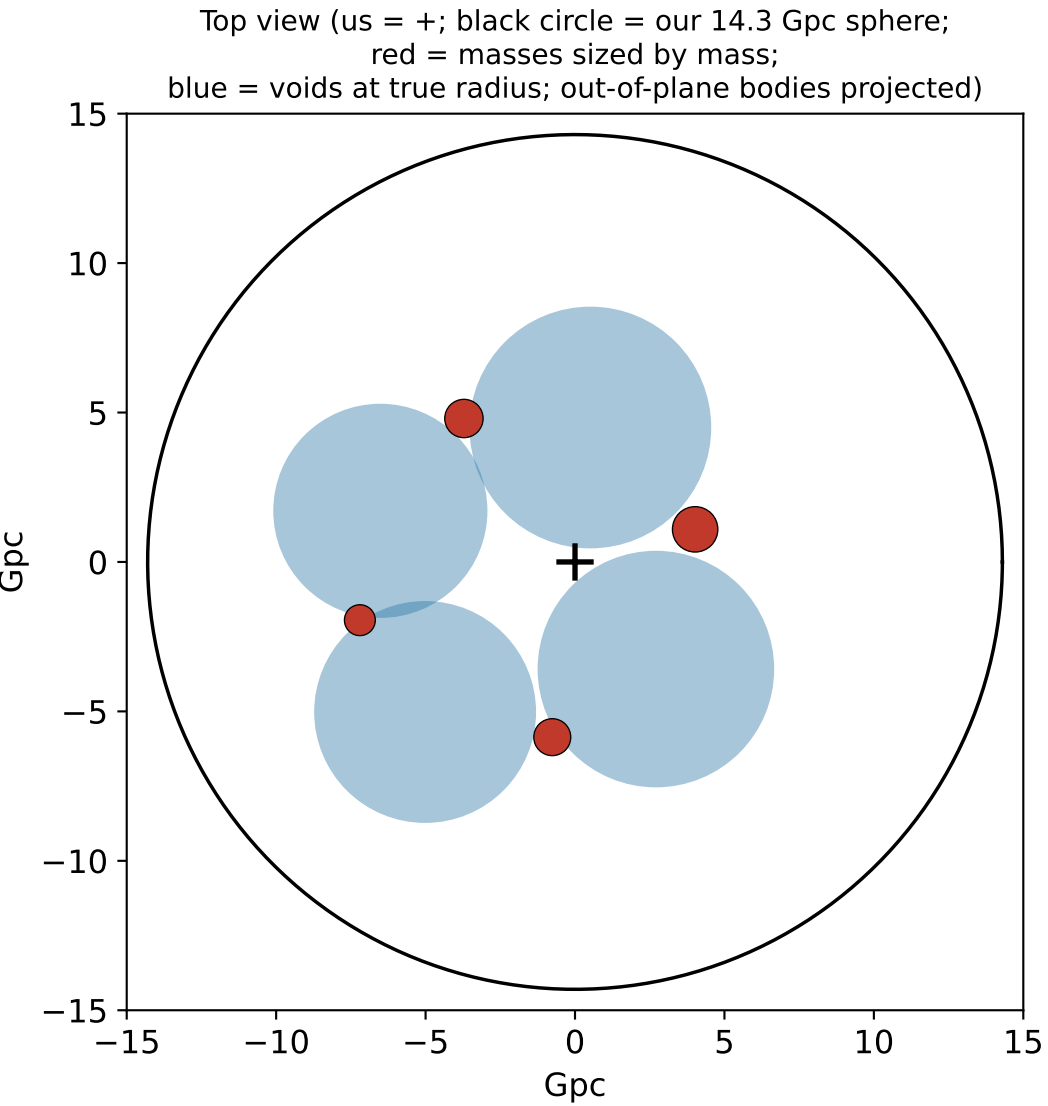


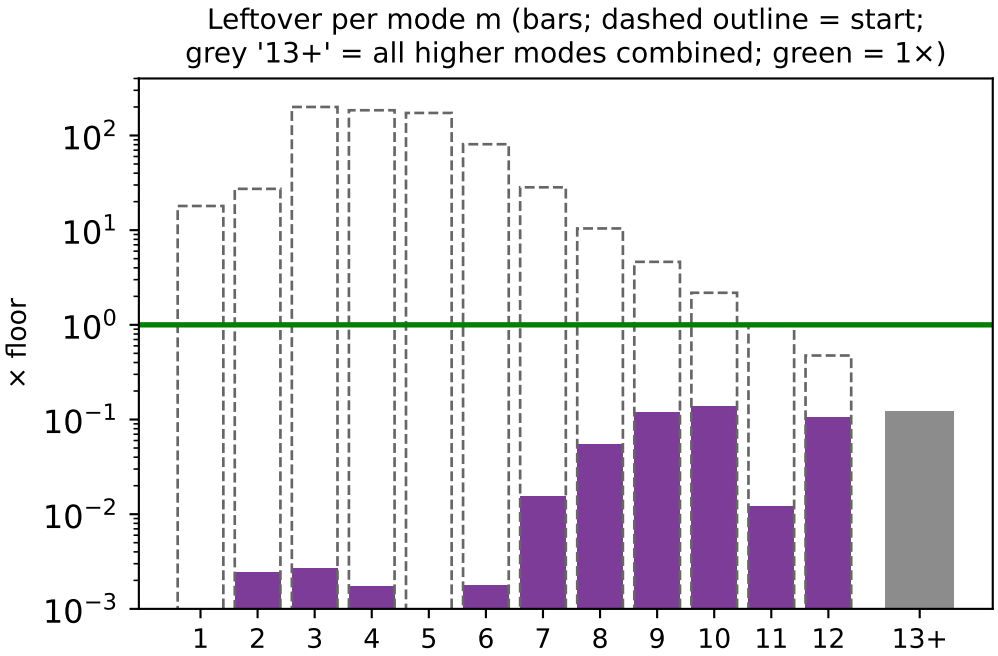


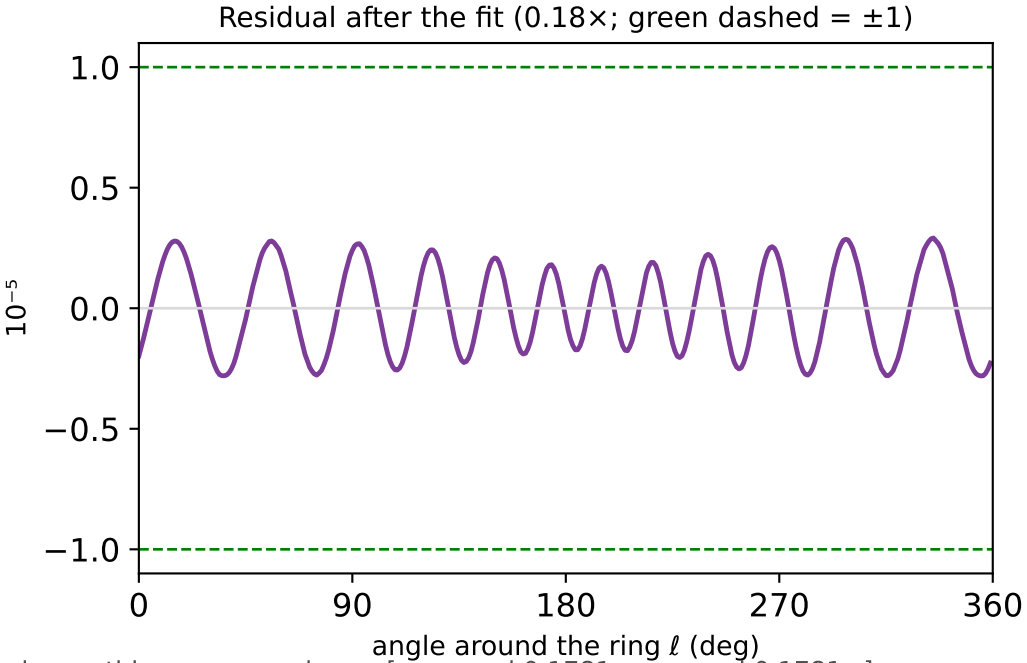


Bodies 0.1–14.2 Gpc from us (never touching the ring); net mass removed smoothly over our sphere [re-scored 0.6206× = saved 0.6206×]

Inside our sphere: masses + deep voids
Example: N = 8 (4 masses + 4 voids), leftover 0.18×

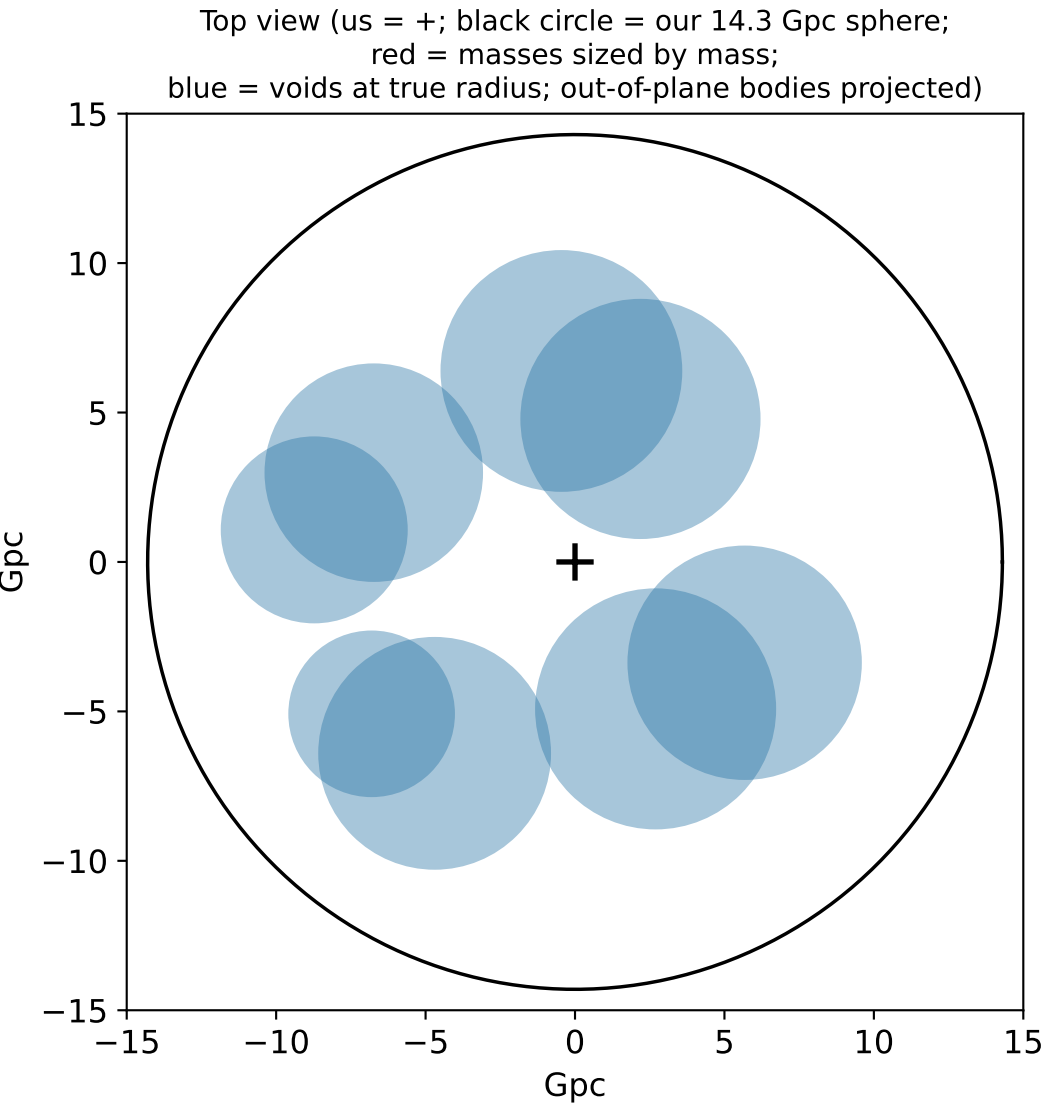


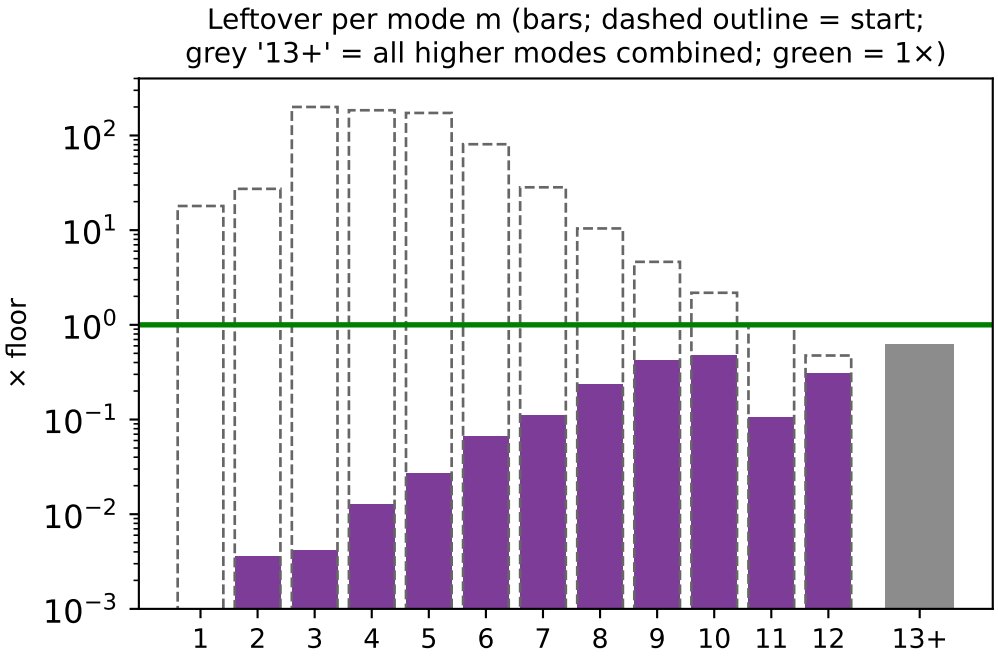


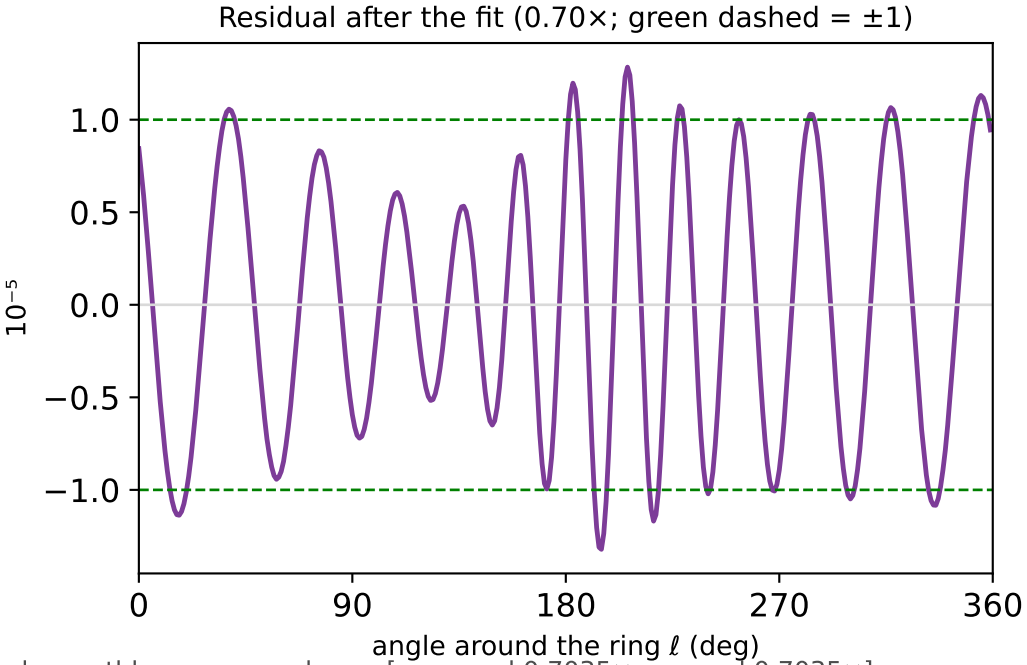


Bodies 0.1–14.2 Gpc from us (never touching the ring); net mass removed smoothly over our sphere [re-scored 0.1781× = saved 0.1781×]

Inside our sphere: deep voids only
Example: N = 8 (8 voids), leftover 0.70×

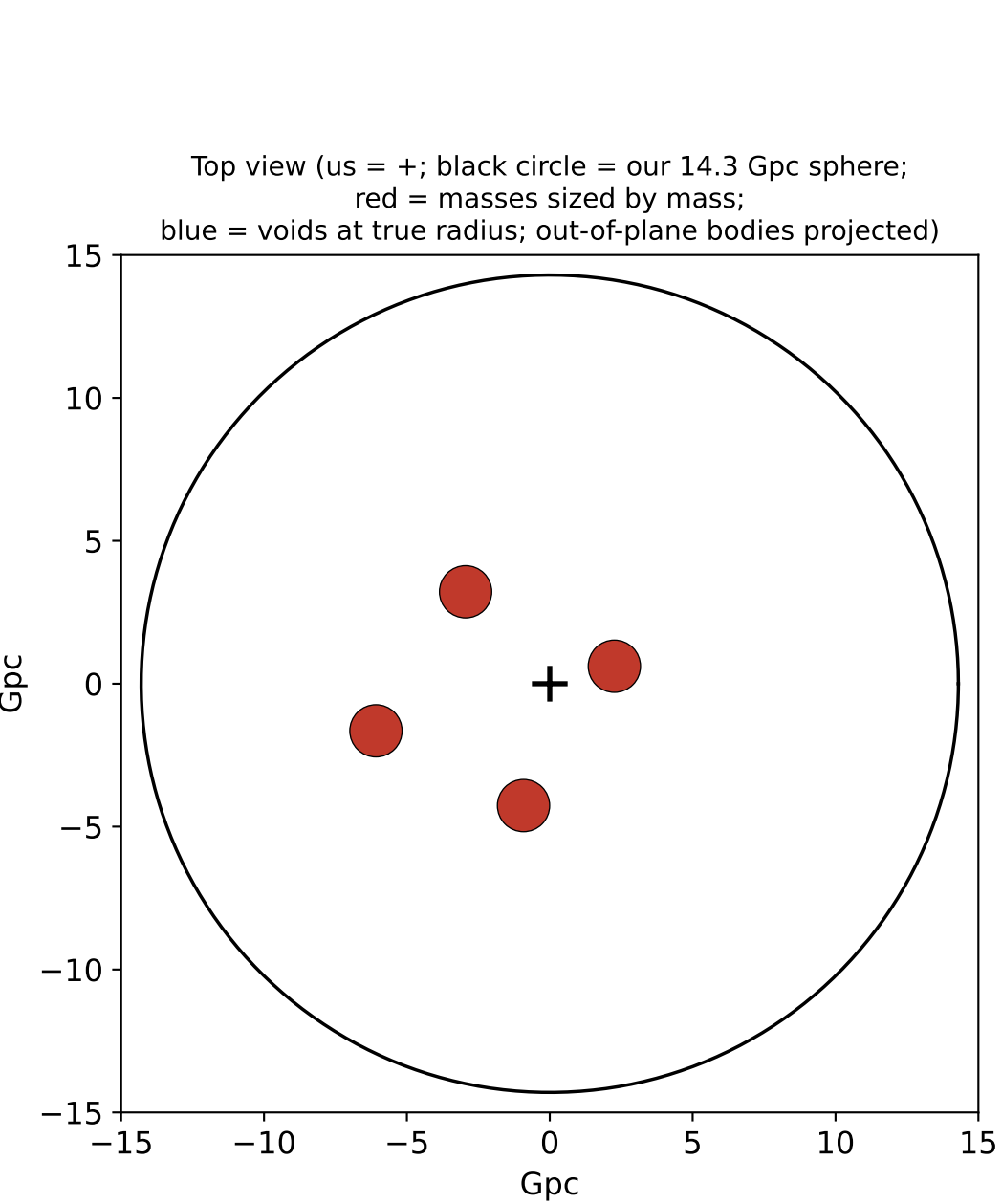


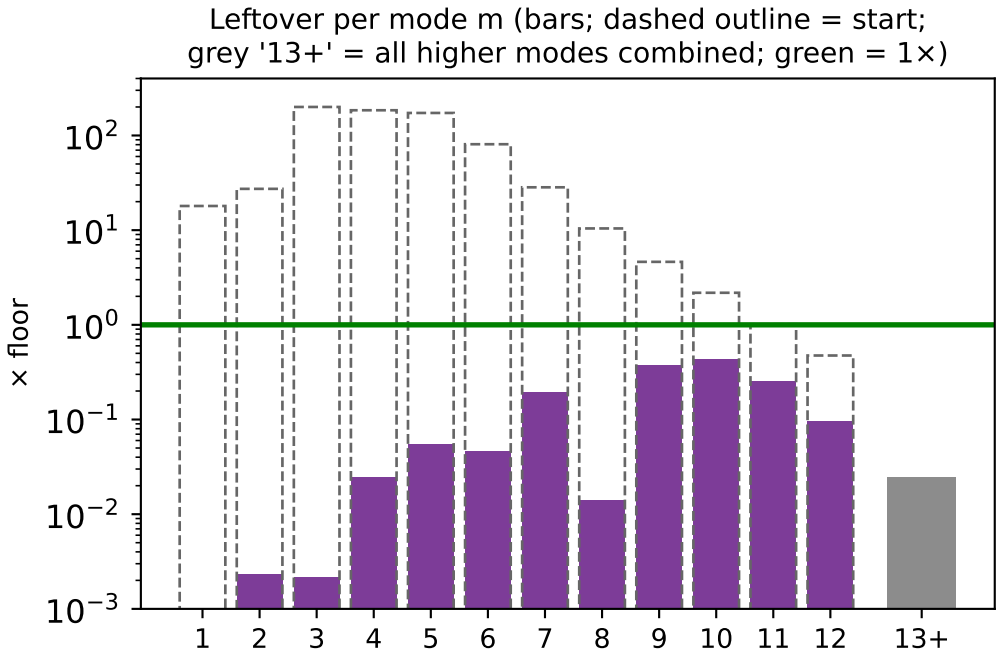


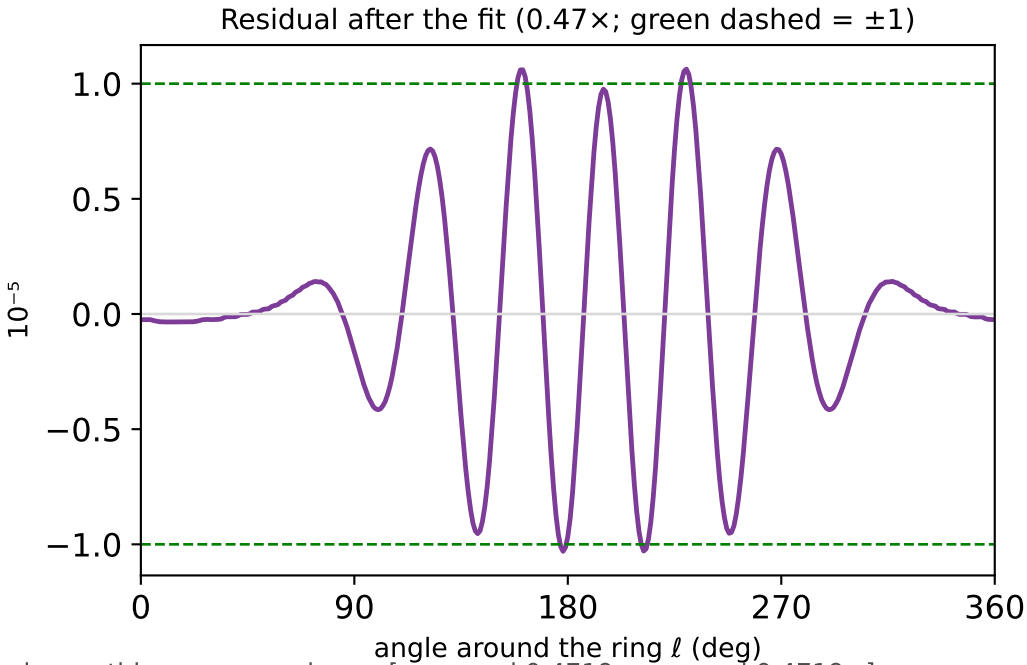


Bodies 0.1–14.2 Gpc from us (never touching the ring); net mass removed smoothly over our sphere [re-scored 0.7035× = saved 0.7035×]

Inside, masses only, cap raised to 1e53 kg
Example: N = 4 (4 masses), leftover 0.47×

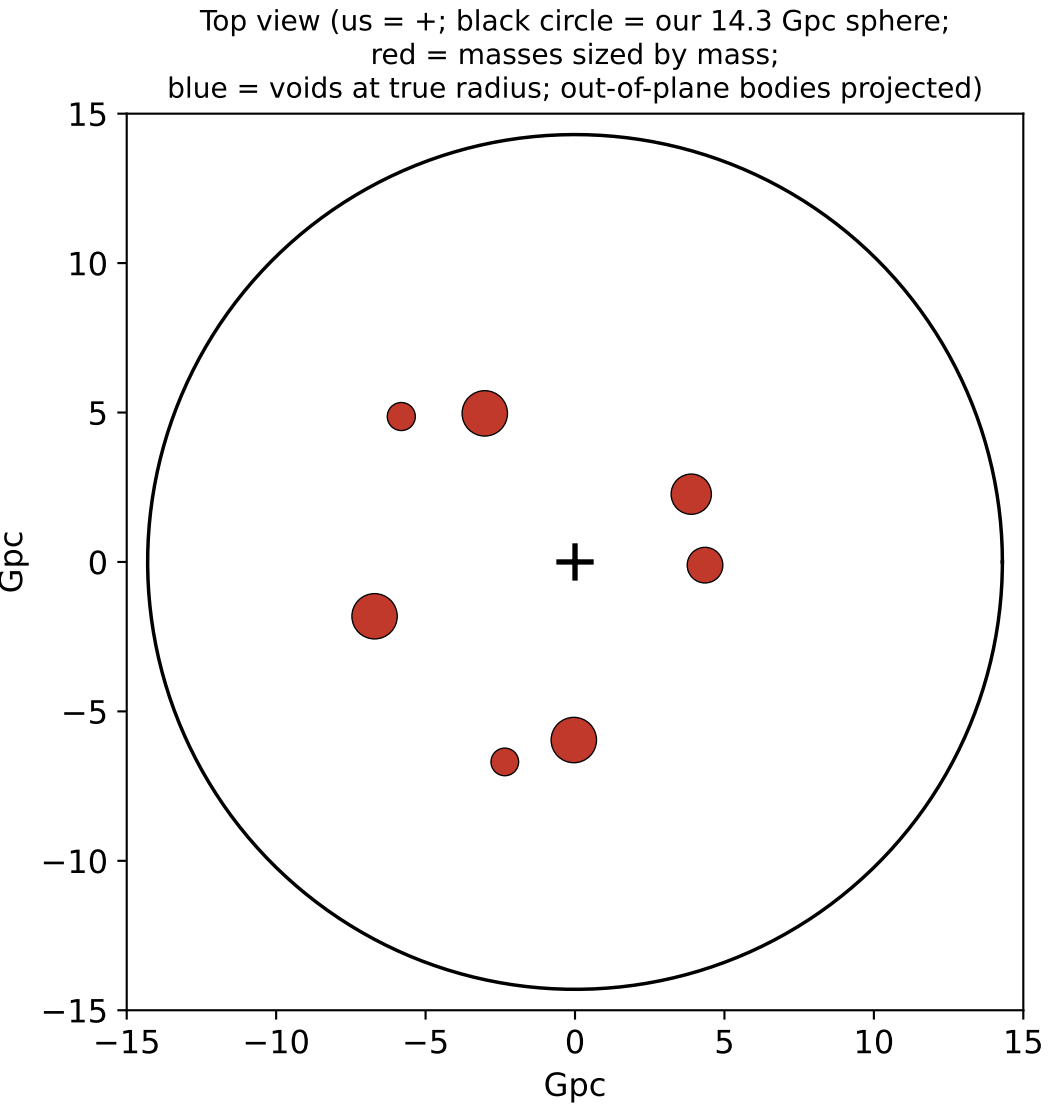


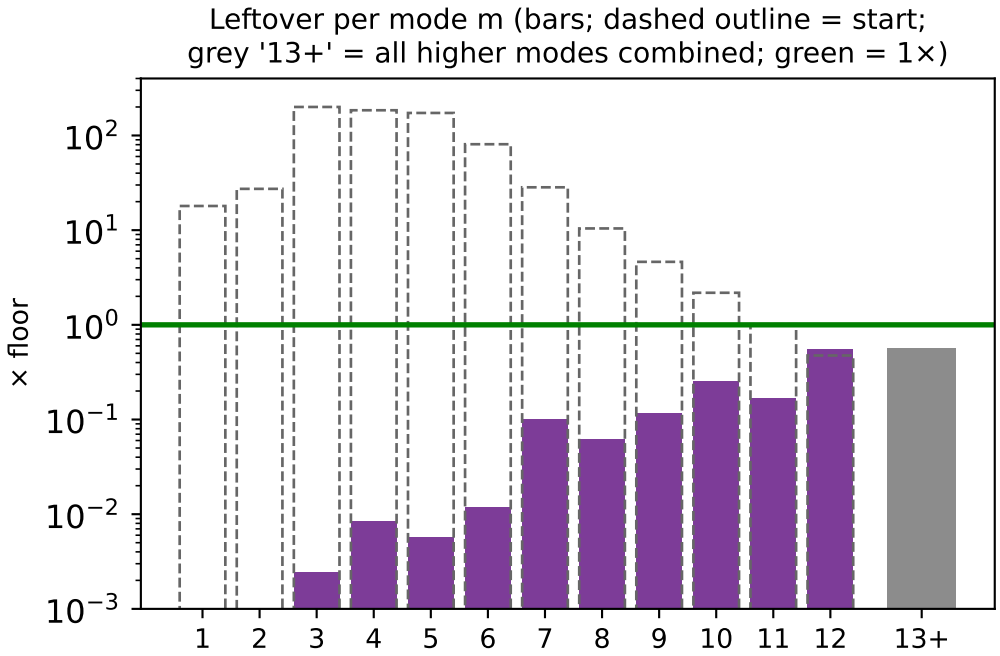


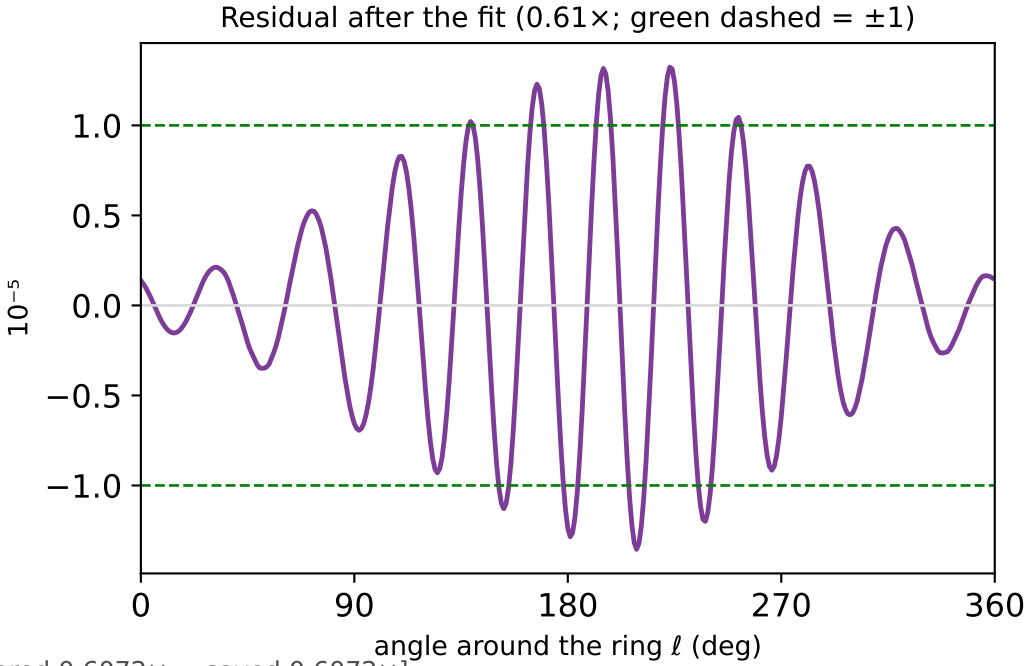


Bodies 0.1–14.2 Gpc from us (never touching the ring); net mass removed smoothly over our sphere [re-scored 0.4719× = saved 0.4719×]

Inside, masses only, no compensation
Example: N = 7 (7 masses), leftover 0.61×

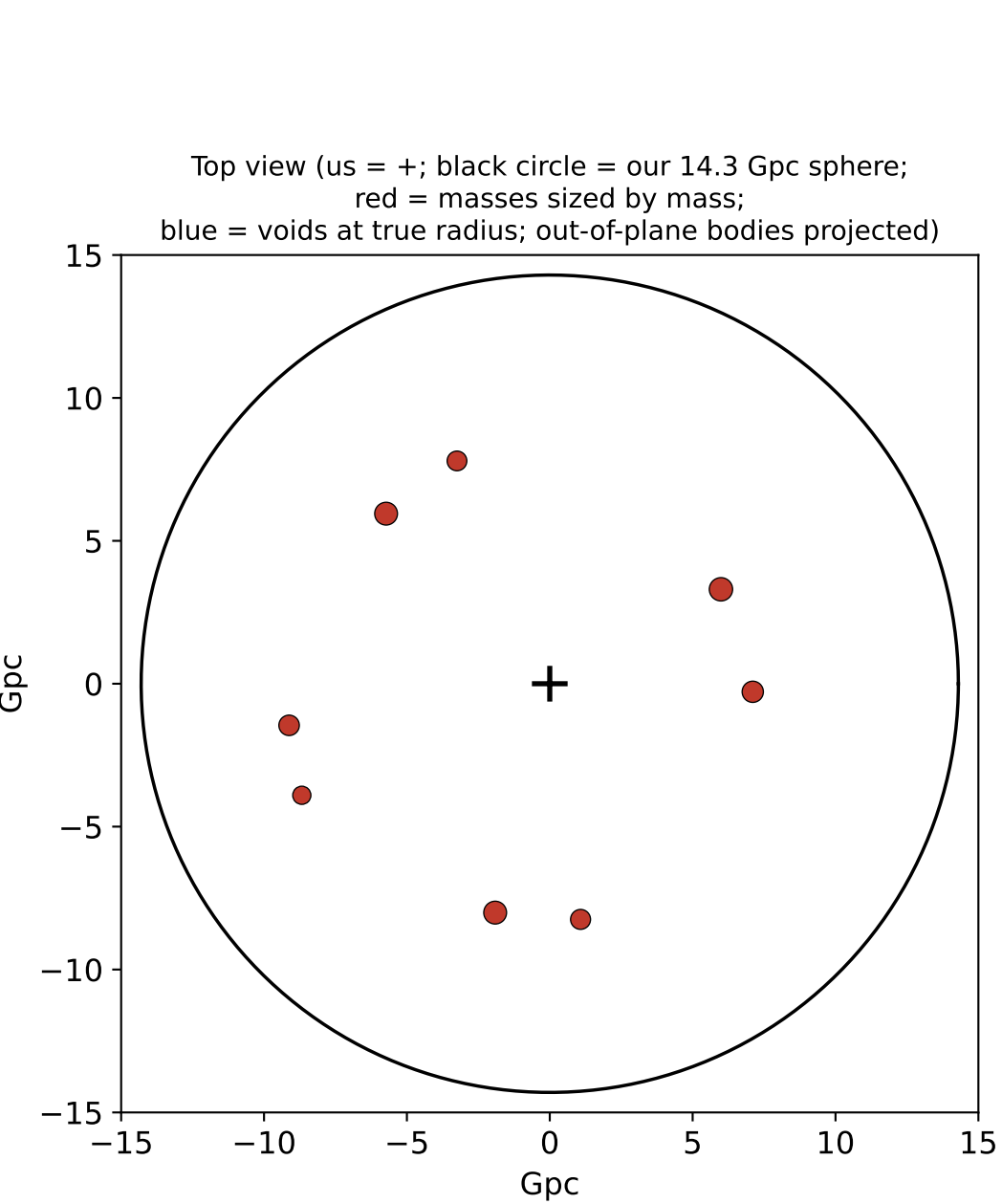


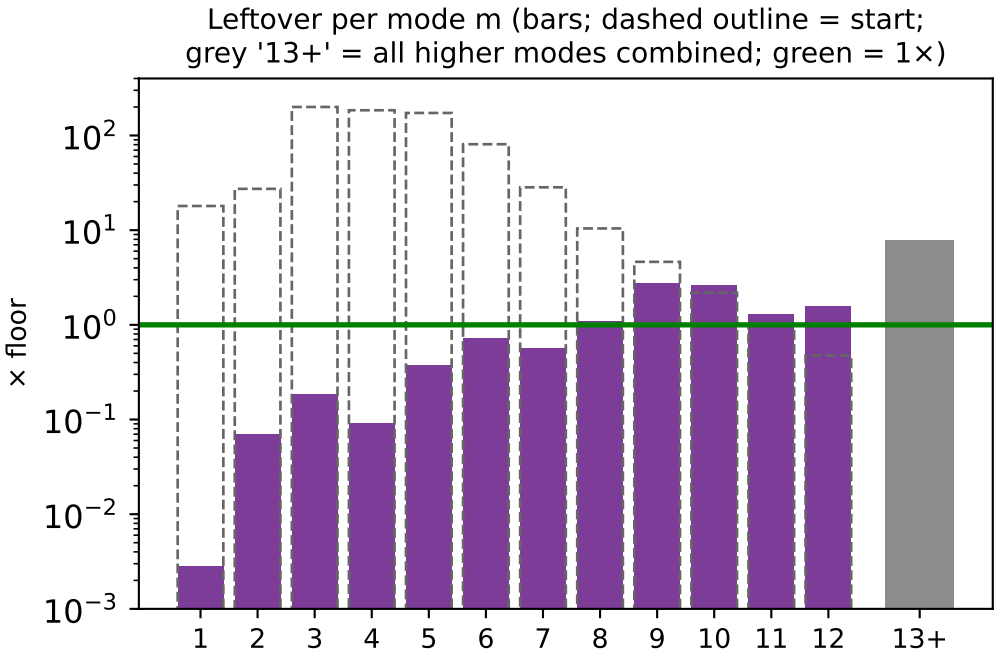


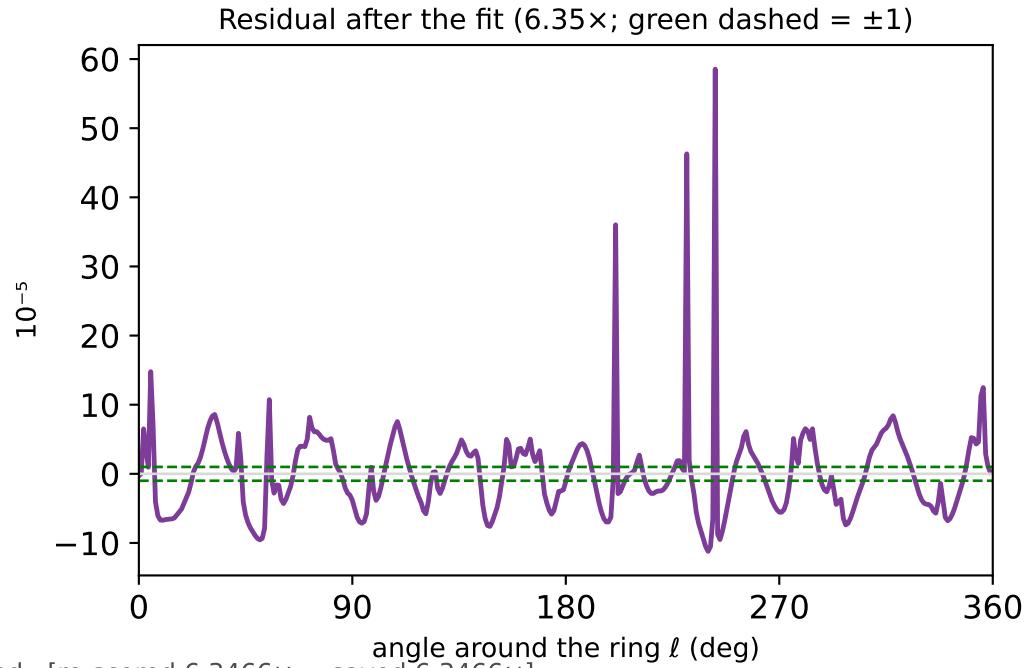


no net-mass compensation [re-scored 0.6072× = saved 0.6072×]

Inside, masses only, compensation taken from outside
Example: N = 8 (8 masses), leftover 6.35×

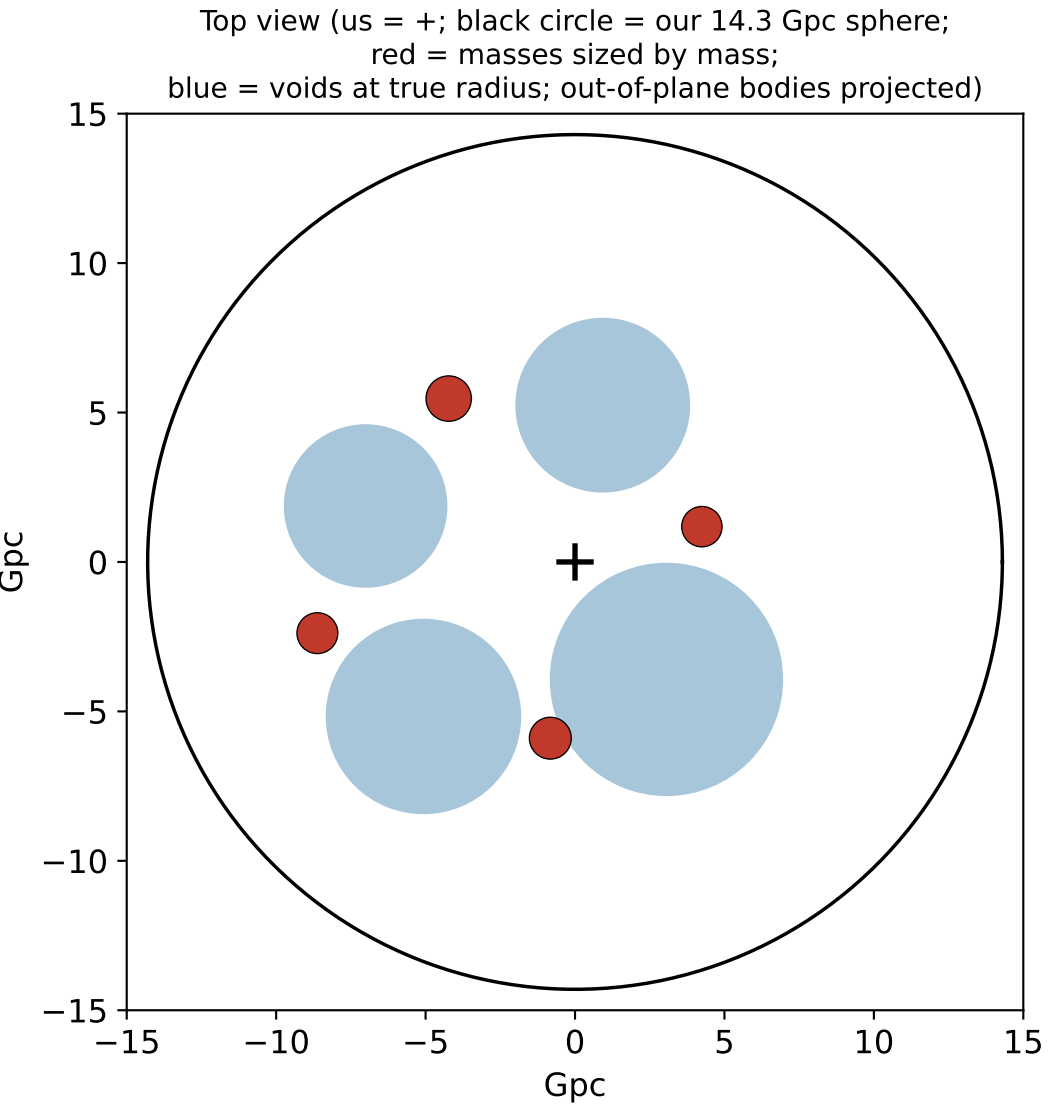


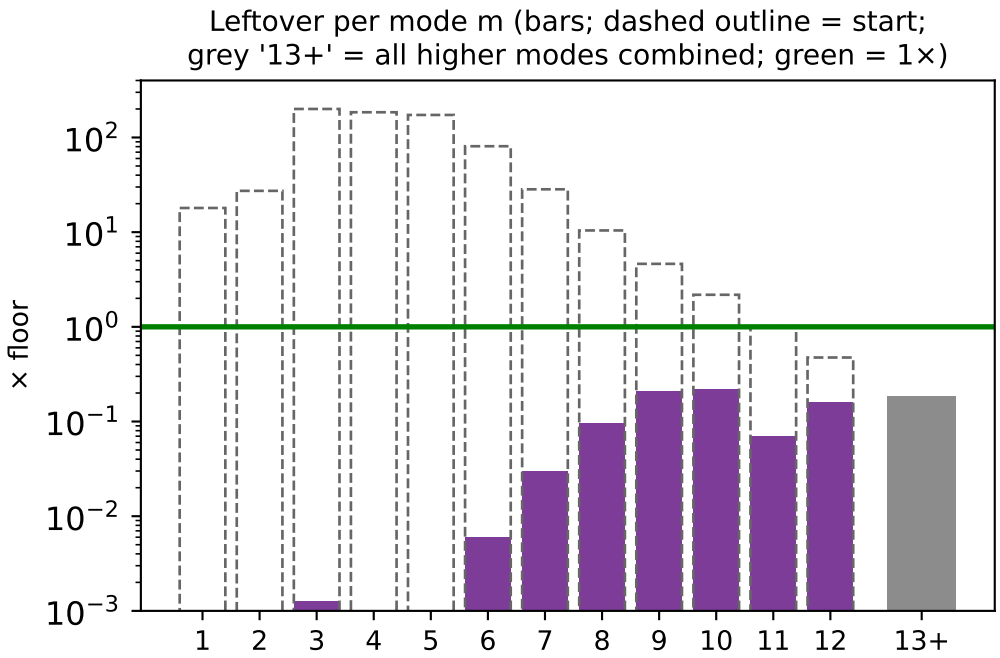


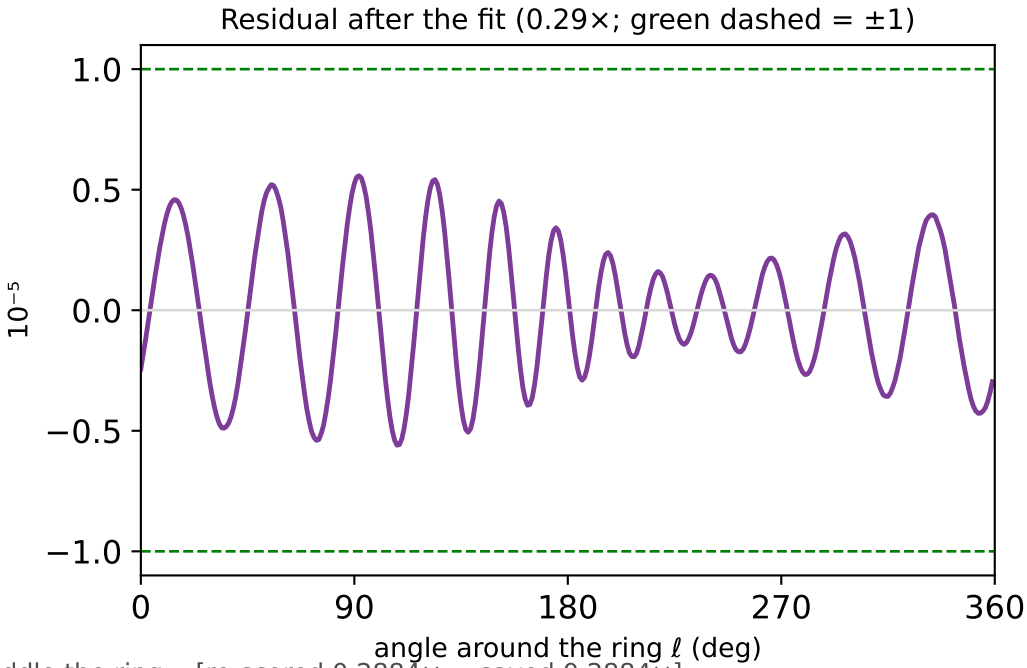


net mass taken from the 14.3–30 Gpc surround [re-scored 6.3466× = saved 6.3466×]

All open (anywhere 0.5–43 Gpc; smooth-density limit)
Example: N = 8 (4 masses + 4 voids), leftover 0.29×

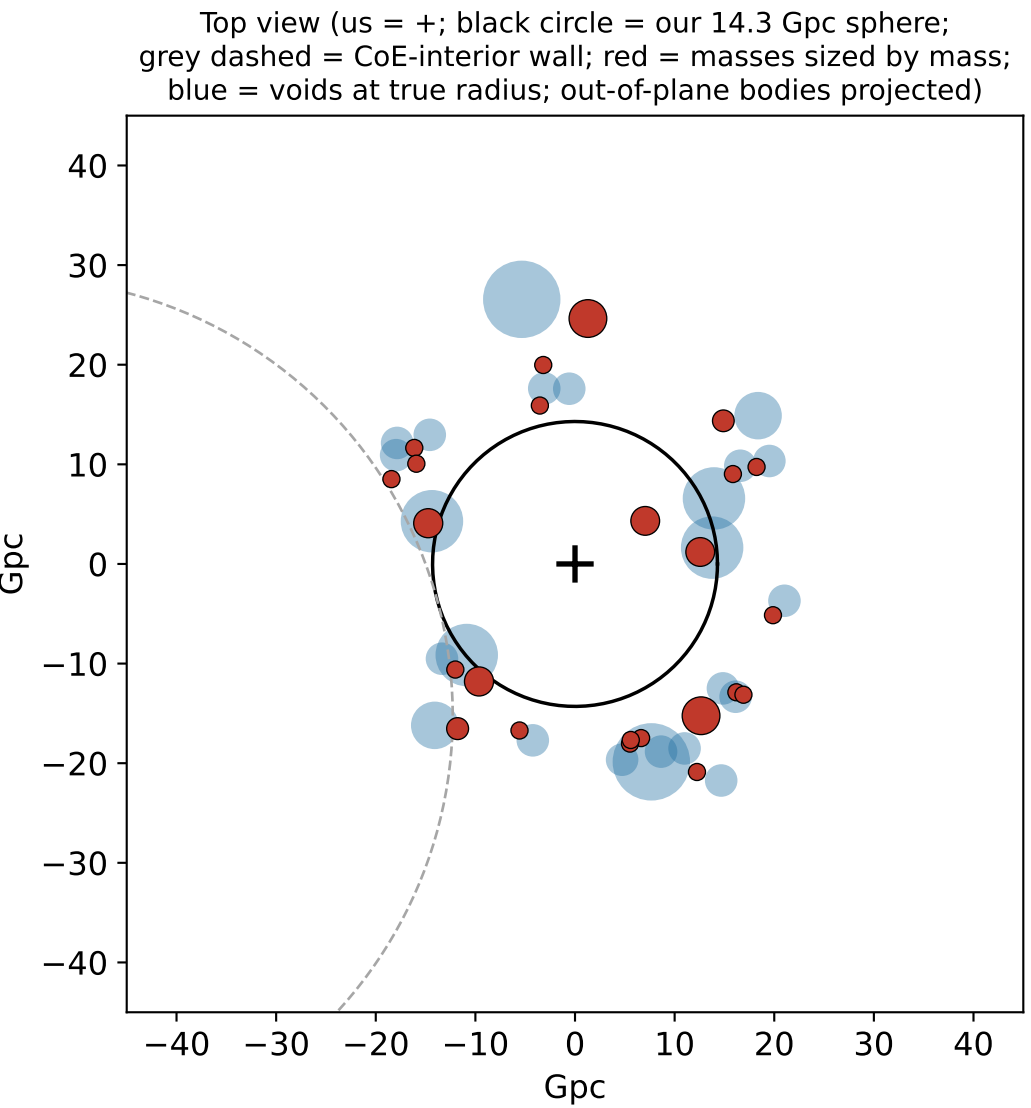


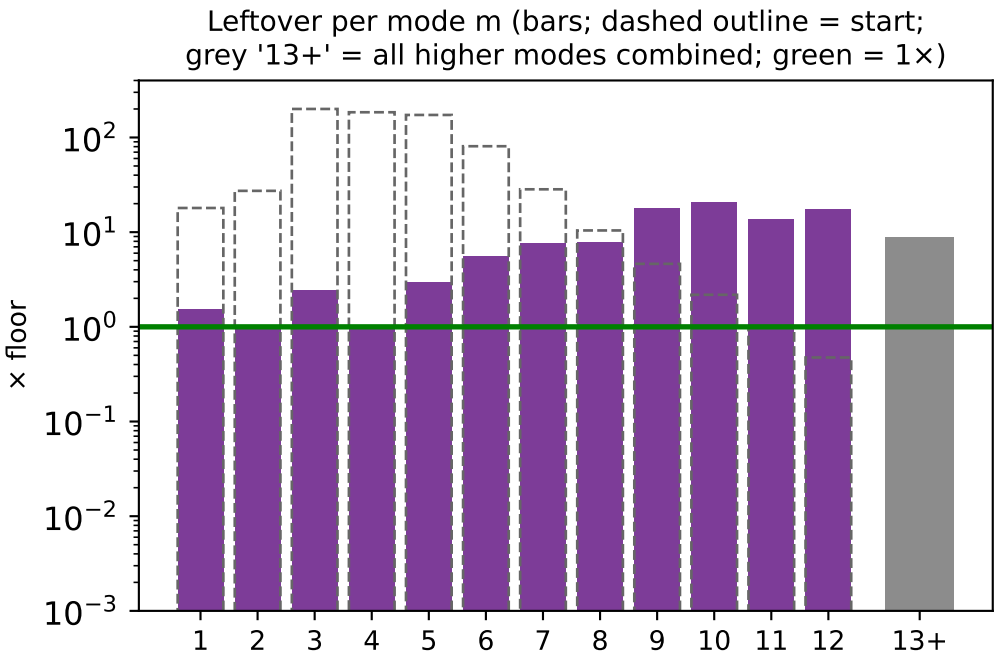


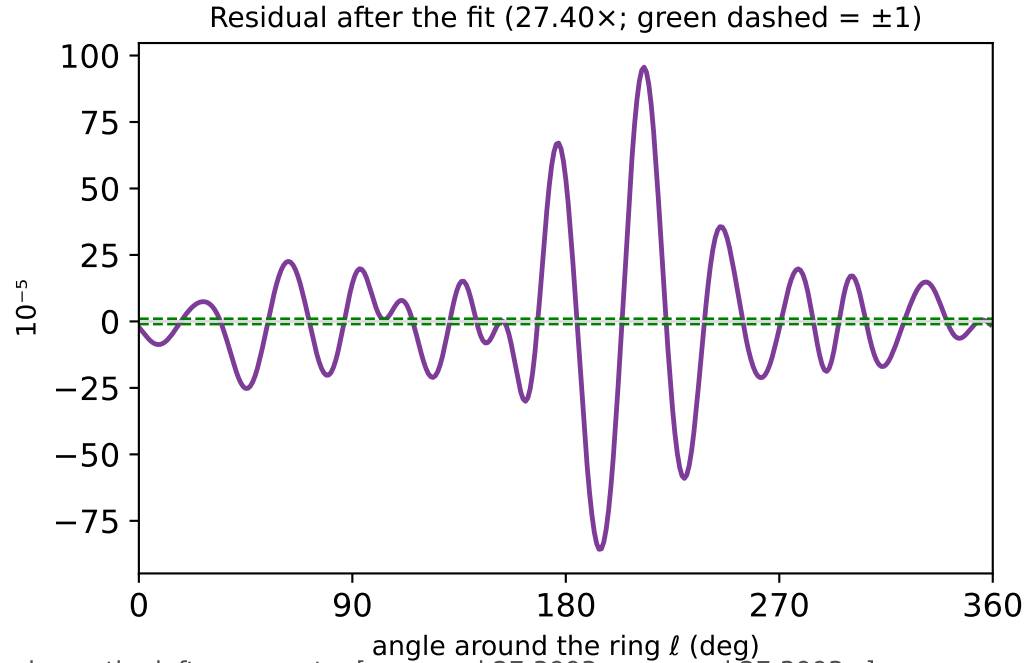


No wall, no visible-sphere exclusion; bodies may not straddle the ring. [re-scored 0.2884× = saved 0.2884×]

Sequential A: near region fixed pair by pair (deep voids + compact clusters)
Example: N = 48 (24 masses + 24 voids), leftover 27.40×

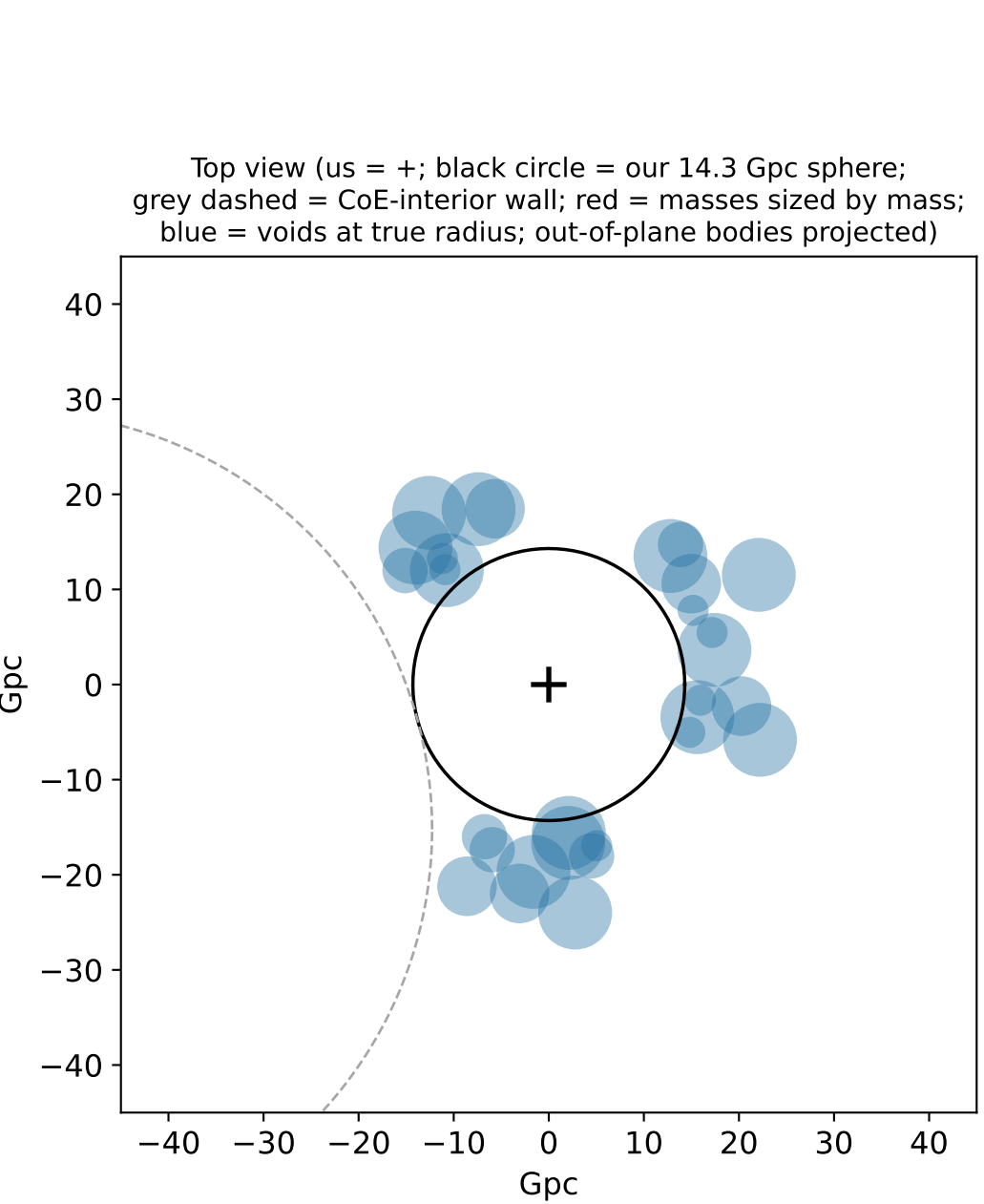


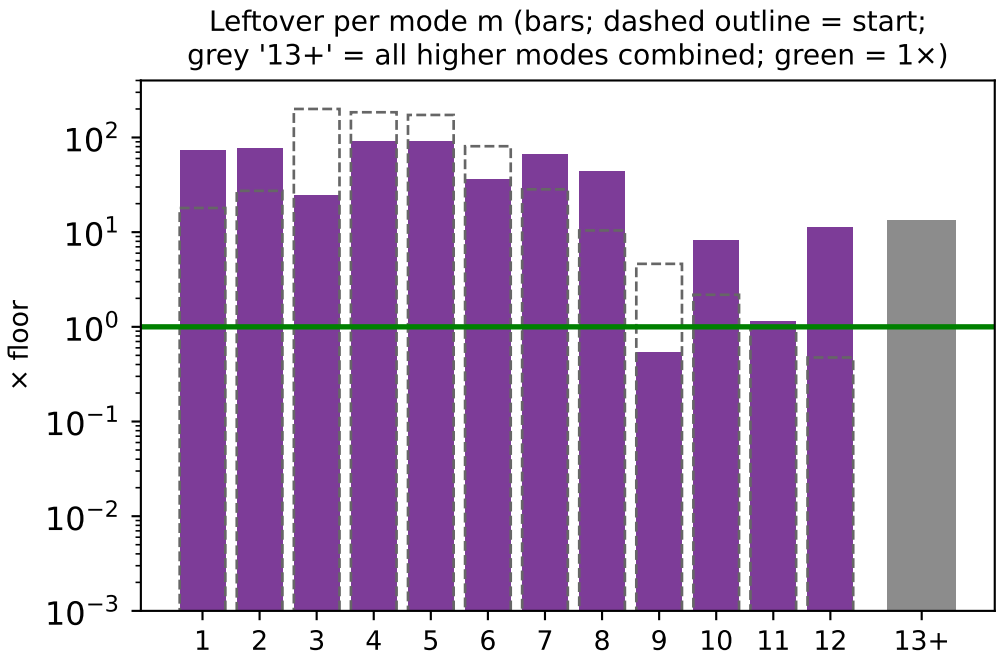




24 void+cluster pairs placed one at a time (14.3–30 Gpc), each chosen to lower the leftover most. [re-scored 27.3993× = saved 27.3993×]

Sequential A: near region fixed, shallow (Clampitt & Jain) voids
Example: N = 30 (30 voids), leftover 135.30×



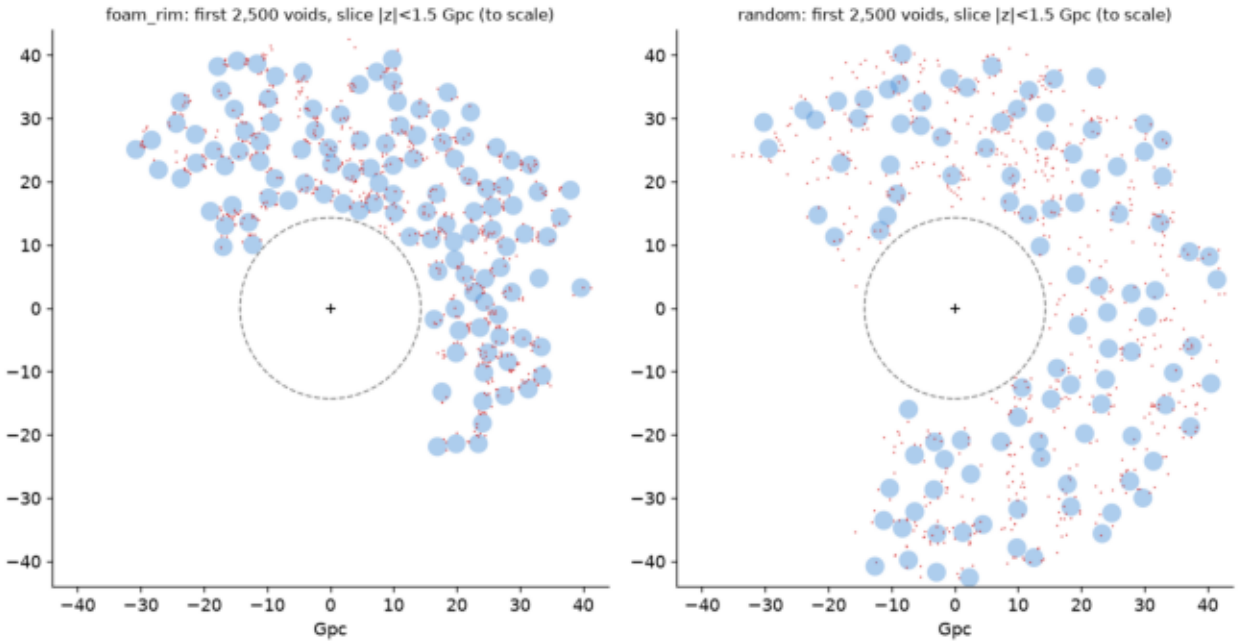


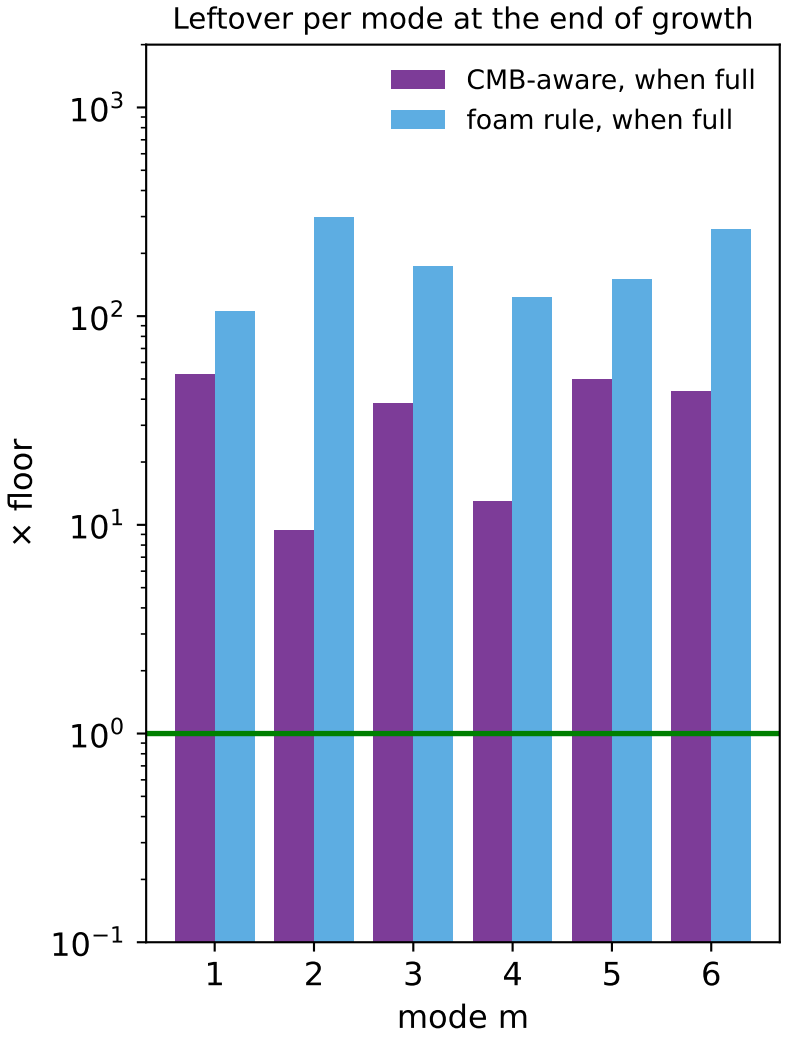


Shallow compensated voids placed one at a time. [re-scored 135.3048× = saved 135.3048×]

Sequential B: grow from empty, one body at a time (example)

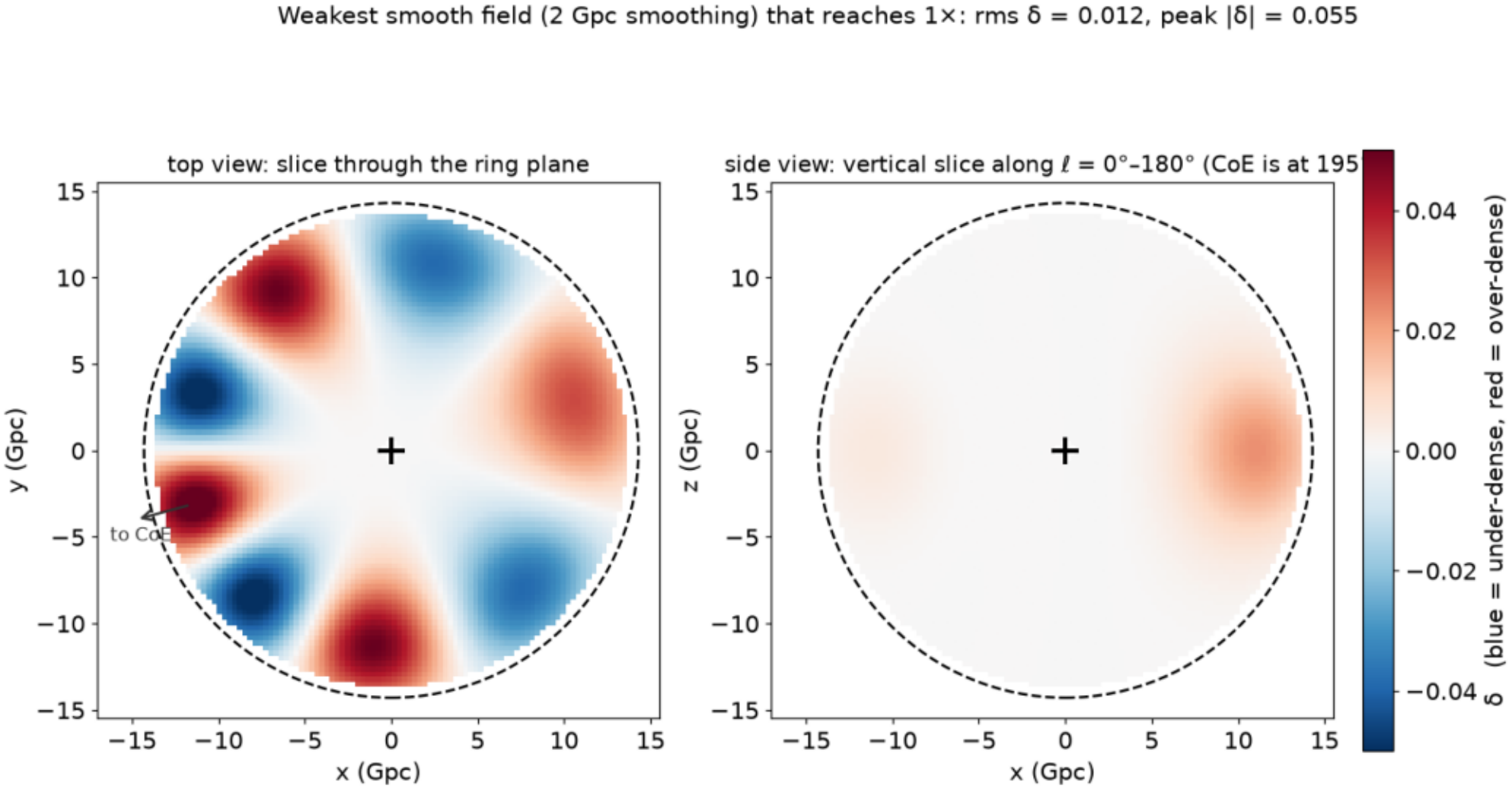
Example grown universes (slices, saved chart)





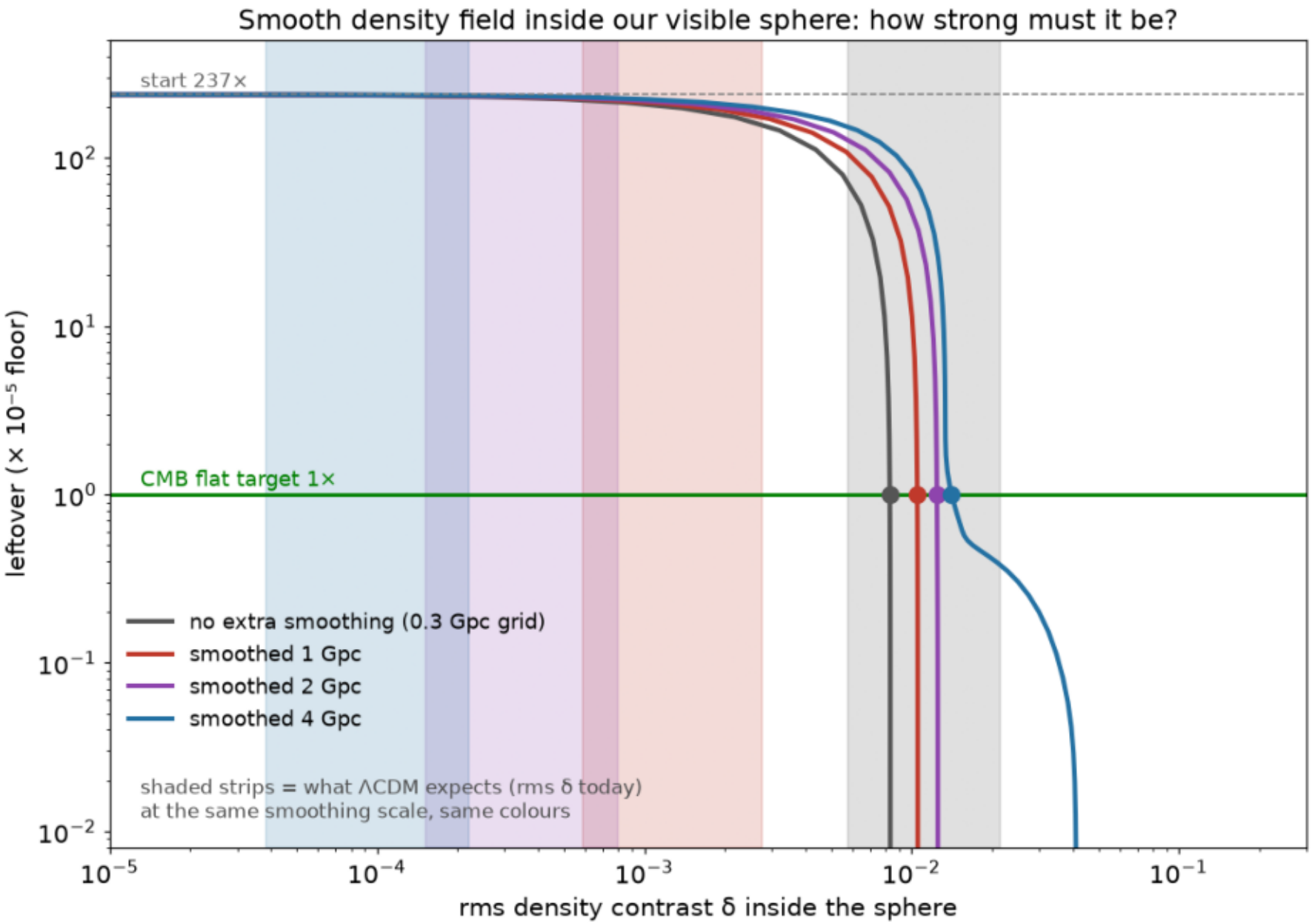
Growth layouts (tens of thousands of bodies) were not saved body-by-body, so no ring residual is re-scored here; the per-mode leftover is the saved final value.

Field-type scenario: smallest smooth density field that reaches 1× (2 Gpc smoothing)



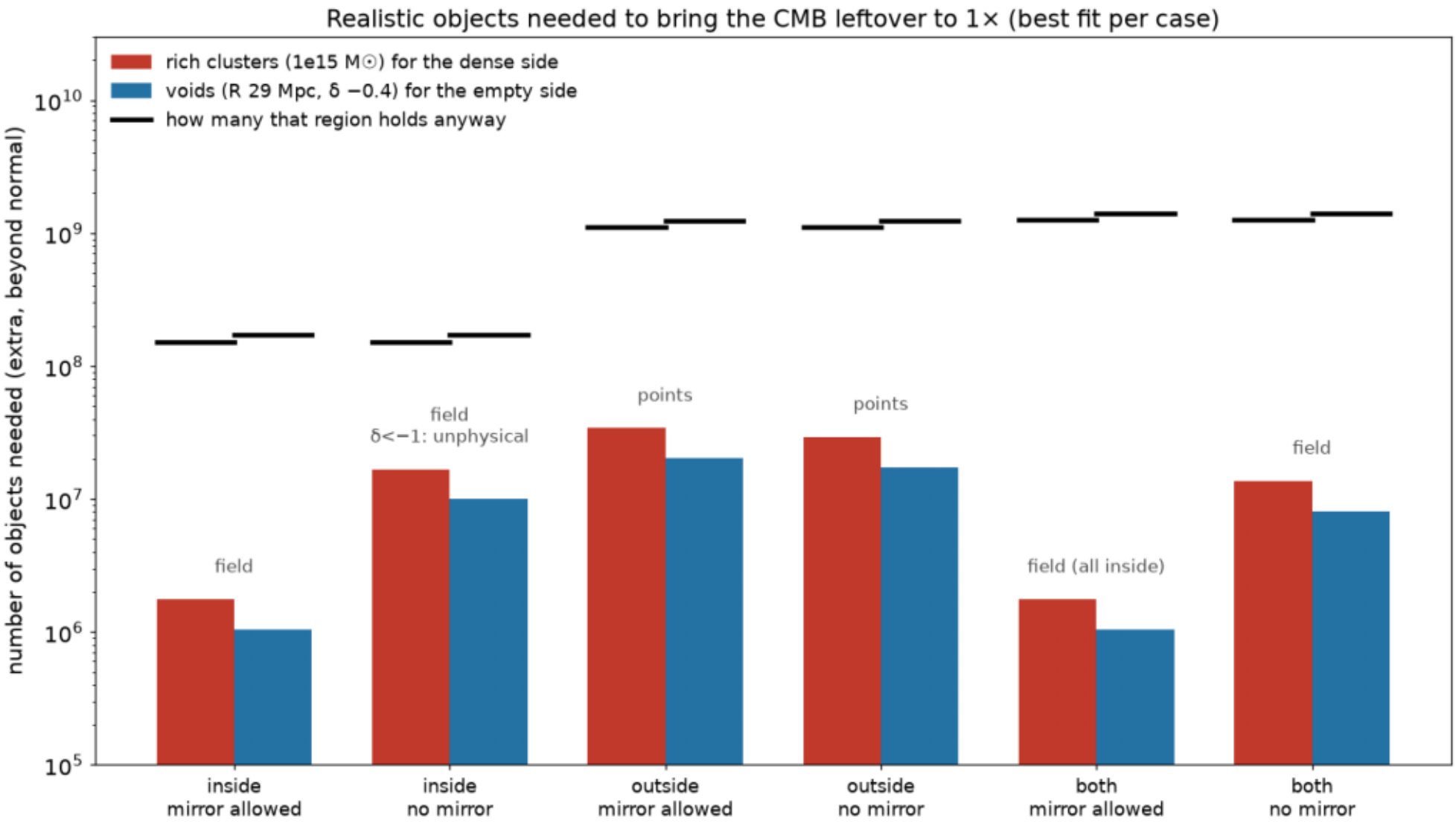
Instead of individual bodies, a smooth density field inside our sphere is solved for directly (net mass zero). Red = slightly overdense, blue = slightly underdense; rms about 1.2%.

Smooth field: leftover vs how strong the field has to be



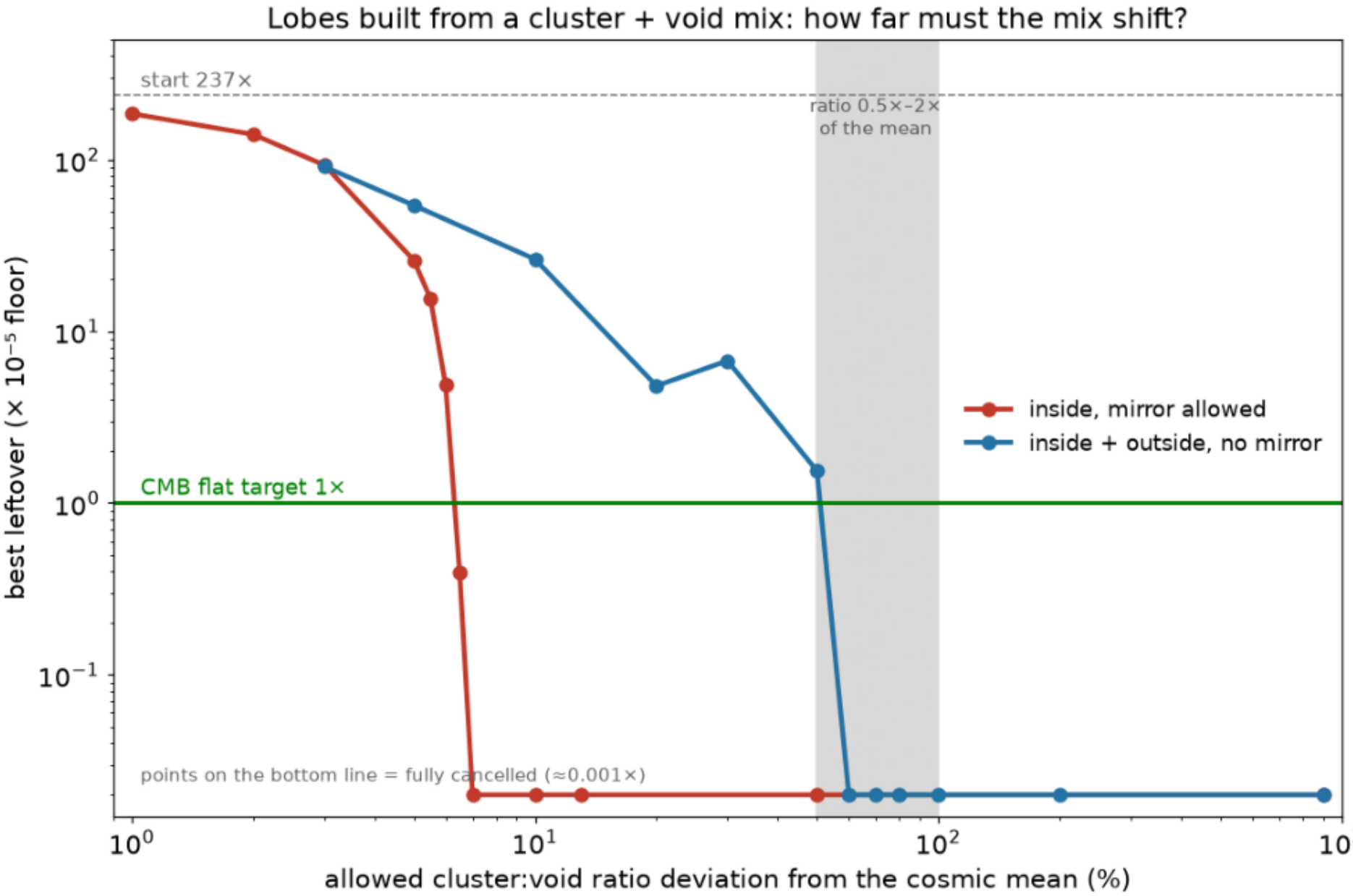
Leftover falls from 237× to 1× as the field's rms density contrast goes from about 0.5% to 1.0–1.4% (depending on smoothing).

Realistic counts: how many extra clusters / voids a 1× solution corresponds to



Counts are 'extra objects beyond normal' (a rearrangement of the normal cosmic web), per scenario, compared with how many each region holds anyway.

Mixed cluster:void ratio: leftover vs how far the local ratio may differ from the cosmic mean



Inside our sphere (mirror allowed) about ±6.5% reaches 1×; inside + outside with no mirror needs somewhere between ±20% and ±60%.

Takeaways: fewest bodies to reach 1× (or best reached)

- outside: masses only: never reaches 1×; best 4.14× at N = 20.
- outside: masses + deep voids: reaches 1× at N = 11 (0.58×); best 0.582× at N = 11.
- outside: deep voids only: never reaches 1×; best 85.3× at N = 8.
- outside: no mirror layouts: reaches 1× at N = 12 (0.73×); best 0.734× at N = 12.
- giant partners (+ ordinary bodies): never reaches 1×; best 1.23× at N = 10.
- outside, no wall: smooth-density limit: reaches 1× at N = 10 (0.73×); best 0.0937× at N = 12.
- outside, no wall: fully open: reaches 1× at N = 10 (0.46×); best 0.0738× at N = 12.
- inside: masses only: reaches 1× at N = 7 (0.62×); best 0.0249× at N = 12.
- inside: masses + deep voids: reaches 1× at N = 8 (0.18×); best 0.178× at N = 8.
- inside: deep voids only: reaches 1× at N = 8 (0.70×); best 0.704× at N = 8.
- inside: masses, cap 1e53 kg: reaches 1× at N = 4 (0.47×); best 0.012× at N = 10.
- inside: masses, no compensation: reaches 1× at N = 7 (0.61×); best 0.0872× at N = 8.
- inside: masses, compensation from outside: never reaches 1×; best 6.35× at N = 8.
- all open (smooth-density limit): reaches 1× at N = 8 (0.29×); best 0.0731× at N = 12.
- A: near fix, deep voids + clusters: never reaches 1×; best 27.4× at N = 48.
- A: near fix, shallow voids: never reaches 1×; best 135× at N = 30.
- B: growth, CMB-aware: never reaches 1×; best 37.2× at N = 44281.
- B: growth, foam rule: never reaches 1×; best 222× at N = 6661.
- B: growth, random: never reaches 1×; best 213× at N = 5051.
- B: growth, CMB-aware, shallow voids: never reaches 1×; best 163× at N = 4235.
- Field-type: a smooth field of ~1–1.4% rms reaches 1×; a mixed cluster:void ratio needs ~±6.5% (inside) or ±20–60% (no mirror). See the preceding four pages.